

Research Paper

The Transminas and Pirenópolis dike swarms: New evidence for the central and South Atlantic Magmatic Provinces in South America



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ABSTRACT

This study reports new geochemical and geochronological data for the Transminas and Pirenópolis dike swarms in Brazil. The Transminas dikes comprise two branches in southeastern Brazil: The Az-170 swarm, trending NNW–SSE and extending for over 800 km, and the Az-125 swarm, trending NW–SE with a total length of 1100 km. The Pirenópolis dikes, in turn, encompass two NNW–SSE-oriented subswarms (~50 km) in Central Brazil. The dikes are subalkaline/tholeiitic basalts and basaltic andesites with high-Ti ($\text{TiO}_2 = 4.3\text{--}3.1 \text{ wt.}\%$) and low MgO ($4.93\text{--}3.34 \text{ wt.}\%$) compositions, except a low-Ti subswarm ($\text{TiO}_2 < 2.0 \text{ wt.}\%$) that exhibits higher MgO content ($>6.5 \text{ wt.}\%$). $^{40}\text{Ar}/^{39}\text{Ar}$ and K–Ar dating of the low-Ti Pirenópolis dikes yields Triassic dates (~210.7–205.5 Ma), linking their emplacement to the Central Atlantic Magmatic Province (CAMP). In contrast, the high-Ti dikes from both the Transminas and Pirenópolis swarms record Early Cretaceous dates (~133–131.5 Ma), coeval with the initial breakup of West Gondwana, and are therefore associated with the Paraná–Etendeka Magmatic Province (PEMP). High-Ti dikes have $^{87}\text{Sr}/^{86}\text{Sr} = 0.71157\text{--}0.70549$ and $\epsilon_{\text{Nd}} = -0.9$ to -16.2 . Lead isotopes indicate a signature compatible with an enriched mantle source (EMI; $^{206}\text{Pb}/^{204}\text{Pb} = 18.40\text{--}18.12$). The integrated datasets for the high-Ti tholeiites of the PEMP giant dike swarms and those of the Equatorial Atlantic Magmatic Province (EQUAMP) indicate derivation from compositionally similar or possibly shared mantle source reservoirs. Such a connection further supports the integration of these Early Cretaceous provinces into the South Atlantic Magmatic Province (SAMP). When assessing the spatial distribution of the magmatic remnants of the CAMP and SAMP, we conclude that the Transbrasiliano Lineament acted as a tectonic boundary, modulating the emplacement realm of the two LIPs during the Mesozoic.

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1. Introduction

Large Igneous Provinces (LIPs) represent extensive magmatic events, typically surpassing $1 \times 10^5 \text{ km}^3$ in volume and, in some cases, exceeding $1 \times 10^6 \text{ km}^3$, widely distributed in intraplate settings, both continental and oceanic. They are predominantly

composed of mafic rocks, although felsic and ultramafic types have also been reported. LIPs typically form over geologically short intervals, lasting less than 5 Myr, or through a series of brief, intense pulses, with the total duration potentially extending over several tens of million years (Ernst, 2014). These provinces manifest in various geological contexts, including continental flood basalts, volcanic passive margins, and oceanic plateaus (Coffin and Eldholm, 1994; Ernst, 2014). The transport of magma from its mantle source to the crust occurs through magma plumbing systems, comprising dike swarms, sill complexes, sheets, plugs, magmatic underplates, and other types of intrusions (Ernst et al., 2019). Among these, dike swarms stand out as they crosscut vast continental regions, preserving a continuous and mappable record

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of magmatic activity. In this context, dikes are ideal targets for investigating petrogenesis, constraining emplacement timing, establishing large-scale correlations, and interpreting geodynamical processes (Ernst and Buchan, 2001).

In South America, three Atlantic-related Mesozoic LIPs stand out in the geological record: (1) Central Atlantic Magmatic Province (CAMP); (2) Paraná-Etendeka Magmatic Province (PEMP); and (3) Equatorial Atlantic Magmatic Province (EQUAMP). The CAMP is a tholeiitic LIP linked to the breakup of Pangea at the Triassic-Jurassic boundary, with a broad geographical distribution, spanning parts of present-day North America, Europe, Africa, and South America (Marzoli, 1999; Davies et al., 2017; Marzoli et al., 2018). This wide distribution was likely due to the extensive contiguous continental areas (Pangea) available during its emplacement. The South American CAMP occurrences have been reported in regions of Brazil, Guianas, Venezuela, and Bolivia (Oliveira et al., 2021, 2022). In northeastern Brazil, the CAMP overlaps with the EQUAMP, a tholeiitic plumbing system-type LIP that are hosted in the area adjacent to the geographic region of the Equatorial Atlantic margin (Hollanda et al., 2019). The EQUAMP is related to the breakup of West Gondwana during the Early Cretaceous, ca. 70 Myr after the emplacement of CAMP.

In the south-central region of South America and its southwestern African counterpart, the PEMP forms a significant Early Cretaceous LIP (Peate et al., 1992; Janasi et al., 2011; Pinto et al., 2011; Szatmari and Milani, 2016; Hartmann et al., 2019; Kingsbury et al., 2023), also associated with the West Gondwana disruption. The PEMP is composed of extensive flood basalts, sill complexes, and mafic dike swarms. While predominantly basaltic/tholeiitic, the province also includes silicic volcanic units and (rare) ultramafic rocks (Peate, 1997; Marsh et al., 2001). Additionally, an important swarm of alkaline intrusions occurs pre-, syn-, and post-emplacement of the PEMP's mafic-silicic magmatism (Gomes, 2025). Recent studies have shown that PEMP and EQUAMP exhibit a geochemical-isotopic and temporal similarity (Macêdo Filho and Hollanda, 2022; Macêdo Filho et al., 2023a,b; Oliveira et al., 2025), therefore constituting a single greater LIP, encompassing an estimated envelope area of at least 1×10^7 km² (Macêdo Filho et al., 2023a); The South Atlantic Magmatic Province (SAMP).

Over the past 40 years, there has been a proliferation of studies dedicated to the geochemical characterization of the Paraná flood basalts (e.g., Peate et al., 1992; Peate, 1997; Marsh et al., 2001; Ewart et al., 2004; Macêdo Filho et al., 2019). Research on the intrusive counterpart has marked an increase over the past 20 years, with regional dike swarms analyzed using geochemical approaches (see Macêdo Filho et al., 2023a for a review). Despite significant advancements, geochemical and isotopic studies of the intrusive components of the Paraná region have been focused in regions from the Santa Catarina to the Espírito Santo states in SSE Brazil (e.g., Piccirillo et al., 1990; Renne et al., 1996; Corval et al., 2008; Valente et al., 2009; Guedes et al., 2016; Almeida et al., 2018, 2021; Florisbal et al., 2018; Marques et al., 2018; Santiago et al., 2020; Carvas et al., 2021). Similarly, in Africa, attention has mainly been directed towards the Etendeka region (Namibia), with only a few studies developed in the coastal area of Angola and South Africa (e.g., Marsh et al., 2001; Ewart et al., 2004; Trumbull et al., 2007; Will et al., 2016; Owen-Smith et al., 2021; Kingsbury et al., 2023).

The northern area of the Paraná Province (Fig. 1A, B), despite hosting one of the largest magma plumbing systems within the entire province, remains poorly explored from a petrological and geochronological standpoint. In that region, the Transminas dikes are particularly noteworthy, as they comprise two giant regional swarms (Chaves, 2014; Macêdo Filho et al., 2023a): One oriented NNW-SSE, spanning over 800 km, and another trending NW-SE, with a total length of 1100 km. A smaller set of dikes occurs in

the Pirenópolis region (Goiás, Brazil), oriented in a north-south direction. In this study, we present new whole-rock geochemistry, including major, trace elements, and radiogenic isotopes (Sr, Nd, Pb), as well as mineral chemistry, ⁴⁰Ar/³⁹Ar and K-Ar geochronological analyses of the Transminas and Pirenópolis dikes. The main objective is to characterize the dike swarms from a geochemical perspective, discuss the petrological and geodynamic processes involved in their formation, and establish correlations with South American LIPs associated with the Mesozoic dispersal of supercontinents.

2. Geological setting

The fragmentation of Pangea began at the Triassic-Jurassic boundary, ~201.6 Ma (Davies et al., 2017; Oliveira et al., 2023), when the CAMP was emplaced shortly before the opening of the Central Atlantic Ocean (Marzoli, 1999). The CAMP tholeiites are predominantly Ti-depleted (TiO₂ < 2 wt.%), although high-Ti types (TiO₂ > 2 wt.%) have also been reported in South America and Africa (Marzoli et al., 2018). The breakup of Gondwana started during the Early Jurassic, ~183 Ma, with the separation of West Gondwana from East Gondwana and the emplacement of the Karoo-Ferrar Magmatic Province (Burgess et al., 2015). A similar process was initiated in the Cretaceous, when West Gondwana began to break apart around 133.8 Ma (Almeida et al., 2018; Rocha et al., 2020, 2023; Sun et al., 2024, 2025; Oliveira et al., 2025), culminating in the separation of South American and African continents. This event generated a series of intracontinental rift basins characterized by intense volcanism, supplied by magma plumbing systems of basaltic/tholeiitic, silicic (tholeiitic series), and alkaline compositions, as documented in the Paraná, Etendeka (plus Greater Cederberg-False Bay), and Equatorial Atlantic provinces (Kingsbury et al., 2023; Macêdo Filho et al., 2023a; Oliveira et al., 2025). In the chemical stratigraphy study of the Paraná flood basalts, Peate et al. (1992) identified at least six types of basaltic magmas, comprising three low-Ti (Gramado, Esmeralda, Ribeira) and three high-Ti varieties (Urubici, Pitanga, Paranapanema). In Etendeka, Marsh et al. (2001) reported seven low-Ti tholeiitic groups (Tafelberg, Kuidas, Horingbaai, Huab, Taefelkop, Albin, and Esmeralda), while only one high-Ti type (Khumib) is identified in the northern part of the LIP. In the EQUAMP, the high-Ti tholeiites are prevalent, with minor occurrences of low-Ti types (Ngonge et al., 2016; Macêdo Filho and Hollanda, 2022; Macêdo Filho et al., 2023b; Dantas et al., 2024).

The specific targets of this study are in the northern part of the Paraná Province (Fig. 1A, B). The Transminas dikes derive their name from their extensive crosscutting of the state of Minas Gerais in southeastern Brazil, as evidenced by magnetometric surveys (Chaves and Correia Neves, 2005). ⁴⁰Ar/³⁹Ar ages of some of these dikes in central Minas Gerais are Cretaceous (Rosset et al., 2007). The dikes intrude Precambrian basement rocks belonging to the Mantiqueira and Tocantins provinces, as well as the São Francisco Craton (Coelho and Chaves, 2016). The Mantiqueira and Tocantins provinces represent extensive Neoproterozoic orogenic belts along the margins of the São Francisco Craton, shaped during the Brasiliano/Pan-African orogeny (Heilbron et al., 2025, for a review). Geographically, the dikes intersect the Mantiqueira Province at their southeastern extent and cross the Tocantins belt in the northwest, with a considerable number of dikes also emplaced along the southern boundaries of the São Francisco Craton (Fig. 1B).

The Transminas dikes stand out as two major giant dike swarms (Fig. 1A): (1) The Azimuth 125° swarm is a series of magnetic anomalies (1100 km-long; Fig. 1B) stretching from the southern region of Minas Gerais State to the western part of Goiás State in the central-western region of Brazil (Coelho and Chaves, 2017;

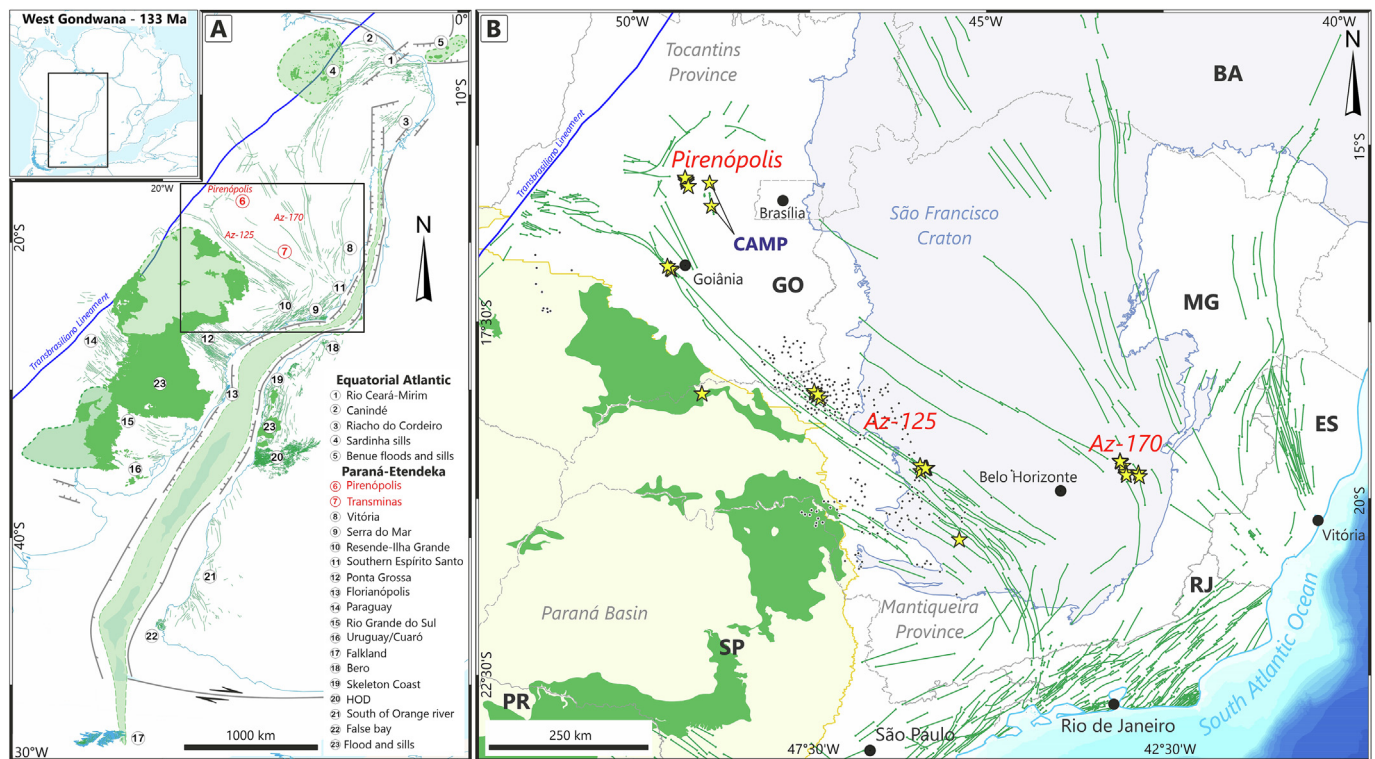


Fig. 1. Reconstruction of West Gondwana showing (A) the Early Cretaceous tholeiitic magmatism of the South Atlantic Magmatic Province – SAMP (modified from Macêdo Filho et al., 2023a). (B) Simplified map of Southeast Brazil focusing on Cretaceous tholeiitic magmatic activity. The Pirenópolis and Transminas dikes are highlighted. The Transminas dikes are splitted into two main branches: Azimuth 125° (Az-125), and Azimuth 170° (Az-170). The yellow stars represent sampling sites of dikes. The black dots represent Mesozoic-Cenozoic alkaline, carbonatitic, and kimberlitic intrusions (Brazilian Geological Survey database). The regional dike trends were mapped with the aid of airborne geophysical data. Full name of Brazilian states: BA, Bahia; ES, Espírito Santo; GO, Goiás; MG, Minas Gerais; PR, Paraná; RJ, Rio de Janeiro; SP, São Paulo.

Rocha et al., 2020), following the Alto Paranaíba Lineament (Gomes, 2025); (2) The Azimuth 170° extends from southeastern Minas Gerais State to central-southern Bahia State (800 km-long; Fig. 1B). To simplify, the dikes associated with the Azimuth 125° and 170° branches will be hereafter referred to as Az-125 and Az-170 dikes (Fig. 1B), respectively. When the Az-125 swarm reaches the Precambrian basement of the Tocantins Province in Goiás (central Brazil), the dikes exhibit a diffuse pattern with NNW-SSE and NE-SW orientations, forming a “splay” towards the Transbrasiliano Lineament (Fig. 1A, B). Finally, the Pirenópolis dikes consist of two NNW-oriented subswarms, each extending for more than 50 km, in central Brazil (in the Pirenópolis town area).

3. Methods

3.1. Sampling

The Transminas swarms extend across a vast transcontinental expanse. To comprehensively survey these expansive swarms, we conducted two field sampling campaigns. The initial campaign spanned approximately 3000 km, traversing the states of Minas Gerais and Espírito Santo. The second campaign encompassed around 4000 km, covering regions within Minas Gerais and extending into the state of Goiás in central Brazil.

The sampling areas were initially delineated based on airborne geophysical surveys, offering a comprehensive regional perspective on the mafic dike swarm. However, the practical execution of fieldwork was significantly constrained by extensive soil cover. The central and southeastern region of Brazil presents environmental conditions characterized by a distinctly tropical climate

with high annual rainfall. This has led to intense rock weathering, contributing to the development of thick soil profiles and further dense vegetation, which together pose considerable challenges to the direct identification and sampling of dikes. Consequently, the field access to outcrops is largely confined to artificial exposures, such as road cuts (Fig. 2A–C), and natural sections (Fig. 2D–F) along watercourses.

Given these limitations, for the sampling of the Az-170 swarm ($n = 11$), our fieldwork was concentrated in the Nova Era area of Minas Gerais. The sampling for the Az-125 swarm was primarily focused on the Campos Altos area of Minas Gerais ($n = 10$), with additional sampling conducted in the Goiânia region in Goiás ($n = 3$). Although we did not visit the area between these two sampling sites, previous mapping work reports mafic dikes in the sector (Seer et al., 2011), also supporting the continuity of the large Az-125 dike swarm. One sample (DMG15) within the São Francisco Craton belongs to the Precambrian (NW-SE) Forminga Swarm (Caxito et al., 2020). In the Pirenópolis dikes, we extensively surveyed outcrops located in the Pirenópolis region ($n = 8$), central Brazil.

3.2. Analytical techniques

3.2.1. Major and trace elements

Representative samples collected from the studied dikes were analyzed for major and trace elements. The samples were processed at the Sample Processing Laboratory (LTA/Geoanalítica-USP Facility), where they were crushed and powdered in an agate ring mill. The particle size achieved was ~300 mesh. The powdered material was then used for major element analysis, which was carried out using X-ray fluorescence at the X-ray Fluorescence Laboratory (Geoanalítica-USP Facility). The methodology used for

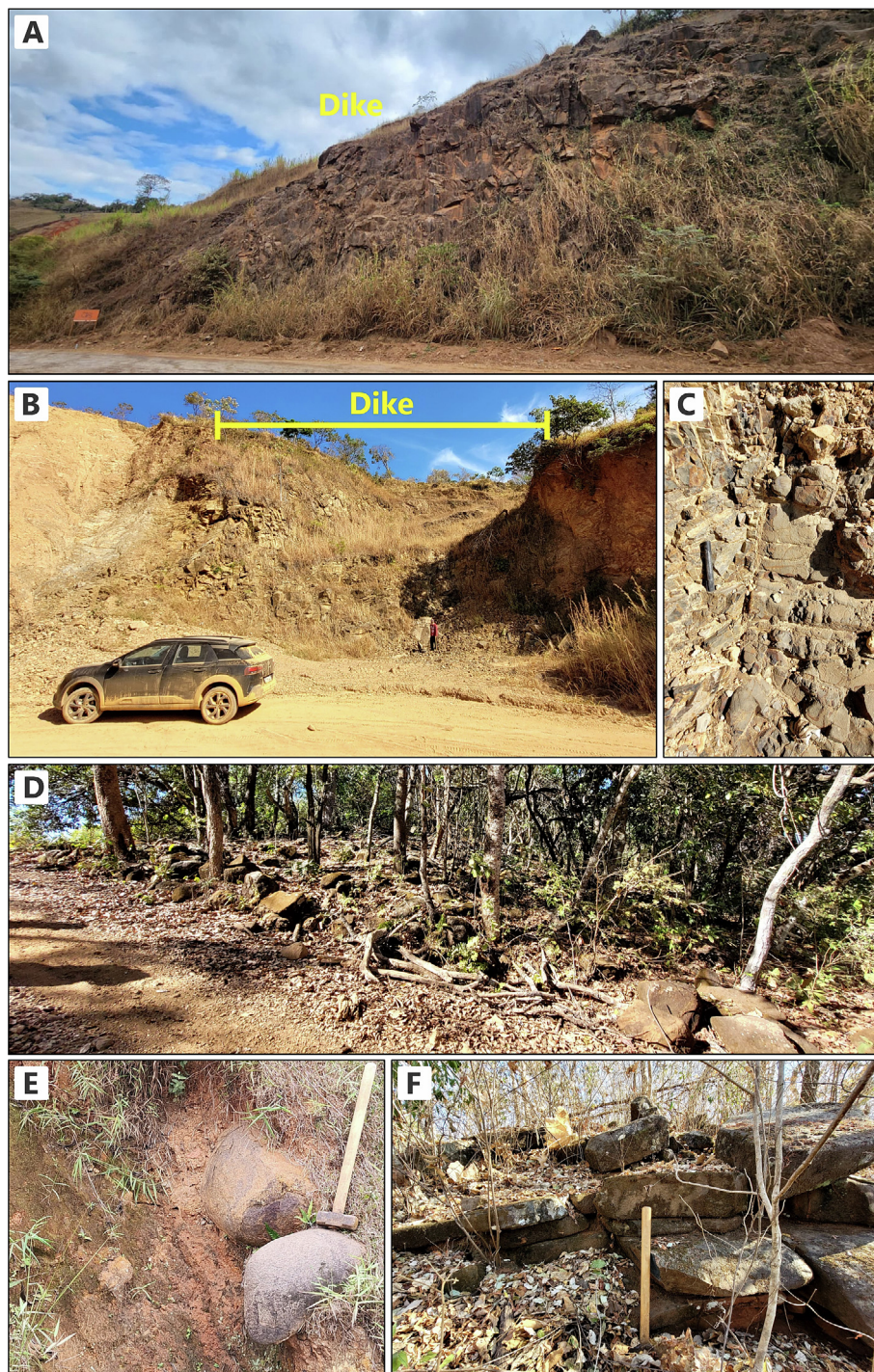


Fig. 2. Field features of the Transminas (Az-125, Az-170) and Pirenópolis dike swarms. (A) View of an outcrop along a road cut of the Az-170 swarm, showing the general alignment and orientation of the dikes. (B) Outcrop in the Az-125 swarm, illustrating the overall geometry and width of the dikes. (C) Close-up of the typical fracturing patterns in the dikes. (D) Outcrop of Pirenópolis swarm dikes, which tend to display positive topography relative to surrounding paraderived metamorphic rocks. (E) An example of erosive features common in the Az-170 dikes is observed as boulders embedded within a reddish soil matrix. (F) Natural outcrop of a Pirenópolis dike swarm showing horizontal disjunction structures, characteristic of this group.

this analysis is outlined in [Mori et al. \(1999\)](#). After the major element analysis, the trace element contents were determined using inductively coupled plasma–mass spectrometry at the ICP Chemistry Laboratory (Geoanalítica-USP Facility). The procedures for this analysis are detailed in [Navarro et al. \(2008\)](#). Additionally, eighteen samples from previous publications ([Rosset et al., 2007](#); [Marques et al., 2016](#); [Coelho and Chaves, 2017](#)) were integrated into our new geochemical dataset to provide a comprehensive

characterization of the studied swarms. The geochemical dataset is provided in [Supplementary Data Table S1](#).

3.2.2. Radiogenic isotopes

We used whole-rock powder samples for isotope analyses of Sr ($^{87}\text{Sr}/^{86}\text{Sr}$) and Pb ($^{206}\text{Pb}/^{204}\text{Pb}$, $^{207}\text{Pb}/^{204}\text{Pb}$, $^{208}\text{Pb}/^{204}\text{Pb}$) at the Isotope Geology Research Center (CPGeo) of the University of São Paulo. The isotope analyses of Nd ($^{143}\text{Nd}/^{144}\text{Nd}$) were done at the

Laboratory of Geodynamic, Geochronological, and Environmental Studies (LEGGA) of the University of Brasília, but two samples were analyzed at CPGeo. The complete dataset is reported in [Supplementary Data Table S1](#).

The process involved dissolving 50 mg of rock powder in a mixture of HF:HNO₃ (3:1), followed by further dissolution in concentrated HCl acid over a period of 5–7 days at sub-boiling temperatures (100 °C). To isolate Pb through ion exchange chromatography, the sample solution underwent an initial treatment on Biorad AG1-X8® resin columns. The eluted matrix solution was then loaded into Sr-Spec® columns to separate Sr and REE (Rare Earth Elements), while Nd and Sm were purified using LN-Spec® columns. The Sr and Pb isotopic analyses were performed using a Thermo Triton Thermal Ionization Mass Spectrometer (TIMS) at the CPGeo/USP. Most Nd isotopic determinations ($n = 22$) were conducted at the Geochronology Laboratory of the University of Brasília (Triton); at the CPGeo/USP, the isotopic determinations of Nd ($n = 2$) were performed using the TIMS – Triton and the ICP-MS Thermo-Neptune.

Total procedure blanks during the analytical program were 114–120 pg for Sr, 32 pg for Nd, and 107–109 pg for Pb. The measured ratios and 2σ uncertainties for standards were as follows: (1) NBS987: $^{87}\text{Sr}/^{86}\text{Sr} = 0.710273 \pm 0.000008$ to 0.710278 ± 0.000006 ; (2) JNdi: $^{143}\text{Nd}/^{144}\text{Nd} = 0.512091 \pm 0.000002$; and (3) NBS981: $^{206}\text{Pb}/^{204}\text{Pb} = 16.900 \pm 0.005$ to 16.903 ± 0.005 , $^{207}\text{Pb}/^{204}\text{Pb} = 15.443 \pm 0.006$ to 15.447 ± 0.006 , and $^{208}\text{Pb}/^{204}\text{Pb} = 36.547 \pm 0.018$ to 36.557 ± 0.019 .

3.2.3. Mineral chemistry

Five representative thin sections of the Transminas and Pirenópolis dikes, each 30 μm thick and carbon-coated, were analyzed to determine the mineral chemistry of pyroxenes, plagioclase, olivine, and Fe-Ti oxides. The analysis was carried out at the GeoAnalítica Lab in the Institute of Geosciences, University of São Paulo, using a JEOL JXA-FE-8530 electron microprobe equipped with five wavelength dispersive spectrometers. The analytical settings included a beam spot size ranging from 5 to 10 μm , a current of 20 nA, and an acceleration voltage of 15 kV. Mineral standards such as anorthoclase, diopside, anorthite, albite, wollastonite, orthoclase, strontianite, fayalite, rutile, chromite, and benitoite were utilized for calibration. Except for Na, which had a counting period of 10 s, all other elements were counted for 20 s. The ZAF correction was done using PsC-EPMA software. The mineral chemistry dataset is available in [Supplementary Data Tables S2–S5](#).

3.2.4. Unspiked K–Ar methodology

To perform the geochronological screening of the mafic dikes, we applied the speedy unspiked K–Ar method according to [Oliveira et al. \(2021\)](#). The mechanical processing of samples involved crushing small whole rock chunks using a steel pestle mill, followed by sieving to obtain a fraction of 16–25 mesh (0.7–1.2 mm). Subsequently, the bulk material underwent washing steps to ensure the thorough removal of dust-sized particles. The K–Ar analyses were performed utilizing a ThermoFisher Scientific Argus VI™ static vacuum mass spectrometer, outfitted with a commercial preparation line (prep-system) specifically designed for argon noble gas purification. The K–Ar dataset is available in [Supplementary Material \(Supplementary Data Table S6\)](#).

3.2.5. $^{40}\text{Ar}/^{39}\text{Ar}$ methodology

Sample preparation and $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology were carried out at the Thermochronology Laboratory of the Isotope Geology Research Center (University of São Paulo, Brazil). Four representative bulk-rock samples were selected for step-heating analysis. Rock fragments were obtained by mechanical crushing in a steel mortar and pestle and subsequently sieved to obtain the 16–25

mesh size fraction (0.7–1.2 mm). This size fraction was cleaned in an ultrasonic bath for 5 min in tap water, followed by a 1-min rinse in distilled water and two additional rinses in distilled water plus ethanol to remove residual fines. Leaching with diluted nitric acid (3% HNO₃) was employed to minimize possible effects of surface alteration and adhering impurities. The leached fragments were dried and loaded into aluminum disks together with the Fish Canyon sanidine (FCs) standard, wrapped in aluminum foil, and encapsulated into a vacuum-sealed silica glass tube for neutron-flux irradiation in the Cd-lined CLICIT facility at the TRIGA® reactor at Oregon State University (USA). Irradiation lasted 10 h.

Step-heating analyses were performed using a CO₂ laser (Fusion 10.6–55 W, Teledyne Instruments, USA) coupled to a multicollector ARGUS VI mass spectrometer (Thermo Scientific, Germany). Argon purification was achieved through an ultra-high-vacuum extraction line equipped with four SAES GP50 getters. Masses 40, 39, and 38 were measured on Faraday detectors with $10^{12} \Omega$ amplifiers, while masses 37 and 36 were measured on $10^{13} \Omega$ amplifiers. A protocol identical to that used for the unknown samples was applied to the FCs standard, which was analyzed as multigrain fractions and used to determine the J value (0.002170 ± 0.000020).

Isotopic ratios were corrected for mass discrimination, background, and procedural blanks using the ArArCalc software ([Koppers, 2002](#)), which was also used for age calculations and graphical output. First age calculation adopted the decay constants of [Steiger and Jäger \(1977\)](#), an atmospheric $^{40}\text{Ar}/^{36}\text{Ar}$ ratio of 298.56 ± 0.31 ([Lee et al., 2006](#)), and an FCs reference age of 28.01 ± 0.04 Ma ([Phillips and Matchan, 2013](#)). Ages initially obtained in ArArCalc were subsequently recalculated in an Excel spreadsheet that performs age and uncertainty propagation using the intercalibration R-factor of [Renne et al. \(2011\)](#). Therefore, the age sets presented in the Results section are those recalculated using the parameters of [Renne et al. \(2011\)](#), with errors quoted at the 95% confidence level, including J -value uncertainty. The complete $^{40}\text{Ar}/^{39}\text{Ar}$ dataset is provided in the [Supplementary Material \(Supplementary Data Table S7\)](#), whereas a summary of the results is presented in [Table 1](#).

Age interpretation followed established criteria: (1) plateau age is defined by at least three consecutive heating steps comprising more than 50% of the cumulative ^{39}Ar released and overlapping within the 95% interval ([Fleck et al., 1977](#)); and (2) when concordant steps were non-consecutive, a weighted average age is calculated.

4. Results

4.1. Magma types

The Transminas dikes display similar chemical characteristics, irrespective of whether they belong to the Az-170 or Az-125 swarms ([Supplementary Data Table S1](#)). This consistency extends to part of Pirenópolis dikes as well. However, Pirenópolis dikes are more fractionated (lower MgO) and have lower Al, Sr, and higher Fe. The high-Ti dikes correspond to subalkaline (tholeiitic series) to transitional basalts, as determined by the Total Alkalis versus Silica (TAS) classification ([Le Maitre, 2002](#)). The composition ranges from basalts to basaltic andesites ([Fig. 3A](#)). Their tholeiitic nature is further supported by a ternary cation plot proposed by [Jensen \(1976\)](#), where samples align along the high-Fe tholeiitic basalt trend (not shown).

The high-Ti tholeiites have TiO₂ = 4.3–3.1 wt.% ([Fig. 3B](#)) with P₂O₅ varying from 1.0 wt.% to 0.4 wt.%, and low LOI (Loss on Ignition) values. These samples exhibit Ti/Y = 886–456, Ti/Zr = 97.7–49.7, La/Yb = 15.3–11.1, and Sr = 1040–453 ppm ([Fig. 3C, D](#)). Comparing the geochemical signature of high-Ti dikes with

Table 1
Summary of the $^{40}\text{Ar}/^{39}\text{Ar}$ geochronological results obtained for the Transminas and Pirenópolis dike swarms.

Sample	Swarm	Magma type	Preferred age $\pm$ 2 s (Ma)	MSWD	$^{39}\text{Ar}\%$	n	$(^{40}\text{Ar}/^{36}\text{Ar})_0 \pm 2$ s (Ma)	Age type	LIP
DMG01	Trasminas (Az-170)	HT	131.49 $\pm$ 1.90	2.1	51	7	433.5 $\pm$ 128.7	Plateau	SAMP
DMG14	Trasminas (Az-125)	HT	133.08 $\pm$ 1.98	2.9	58	10	326.1 $\pm$ 72.0	Plateau	SAMP
DGO01	Pirenópolis	LT	196.39 $\pm$ 2.91	1.5	57	6	302.2 $\pm$ 5.8	WMA	CAMP
DGO02	Pirenópolis	LT	205.5 $\pm$ 3.1; 210.7 $\pm$ 3.4	na	100	20; 13	na	Total fusion	CAMP

Notes: WMA, weighted mean age; na - not applied; HT - high-Ti tholeiites; LT - low-Ti tholeiites; LIP, Large Igneous Province; SAMP, South Atlantic Magmatic Province; CAMP, Central Atlantic Magmatic Province.

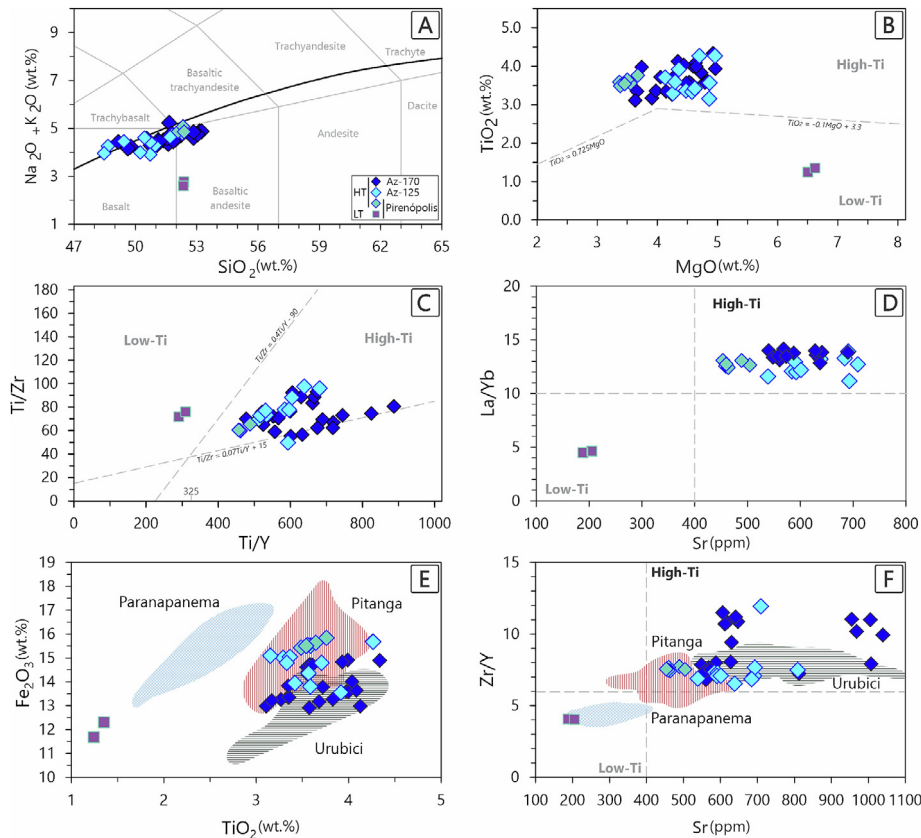


Fig. 3. Geochemical classifications and discriminations for the Transminas (Az-125, Az-170) and Pirenópolis dikes: (A) TAS diagram showing total alkalis versus SiO_2 (wt.%), with compositional fields and dividing line adapted from [Le Maitre \(2002\)](#). (B) Plot of TiO_2 versus MgO (wt.%), (C) Ti/Zr versus Ti/Y ratios, (D) La/Yb ratios versus Sr (ppm). Separation field lines are based on [Macêdo Filho et al. \(2023a\)](#). (E) TiO_2 versus Fe_2O_3 , and (F) Sr versus Zr/Y diagrams used for the discrimination of high-Ti basalt types of the Paraná Province based on [Peate et al. \(1992\)](#).

previous magma groups reported in the Paraná Province ([Peate et al., 1992](#)), the Transminas dikes correspond to high-Ti-Sr magmas that, in some cases, can be classified as Urubici- or Pitanga-type, particularly considering that a significant subset of samples plots analogously to the Pitanga type in TiO_2 vs. Fe_2O_3 plot ([Fig. 3E](#)). On the other hand, the Pirenópolis dikes are classified just as the Pitanga-type ($\text{Sr} < 550$ ppm; [Fig. 3F](#)). A subset of dikes of the Pirenópolis present compositions like basaltic andesites with low TiO_2 (1.3–1.2 wt.%), P_2O_5 (0.14–0.13 wt.%) and high MgO (6.63–6.51 wt.%) contents. They have $\text{Ti}/\text{Y} = 309\text{--}290$, $\text{Ti}/\text{Zr} = 75.9\text{--}71.8$, $\text{La}/\text{Yb} = 4.6\text{--}4.4$, and $\text{Sr} = 205\text{--}188$ ppm ([Fig. 3C, D](#)).

4.2. Petrography and mineral chemistry

The dikes were classified into three textural categories based on the grain size observed in thin sections: fine- (<1 mm), medium- (1–5 mm), or coarse-grained (>5 mm). The petrographic description indicates medium-grained microgabbros, forming hypocrySTALLINE (intergranular) to holocrystalline (subophitic) textures ([Fig. 4A](#)). The mineral assemblage is dominated by plagioclase,

clinopyroxene (augite $\pm$ diopside), olivine, titanomagnetite, ilmenite, sulfides, apatite, and zircon. In the hypocrySTALLINE types ([Fig. 4B, C](#)), the intercrystalline domains are composed of a devitrified matrix and altered glass. Alteration is incipient. Plagioclase crystals can vary in morphology from euhedral to anhedral, exhibiting a lath-like shape ([Fig. 4D, E](#)). Smaller crystals of plagioclase tend to be acicular. Plagioclase typically displays normal zoning, ranging from bytownite to andesine ($\text{An}_{72.7\text{--}33.0}\text{Ab}_{25.8\text{--}53.8}\text{Or}_{1.5\text{--}13.2}$), with ca. 50% of labradorite ($\text{An}_{67.6\text{--}50.0}\text{Ab}_{30.4\text{--}47.1}\text{Or}_{2.0\text{--}2.9}$) ([Fig. 5](#); [Supplementary Data Table S2](#)). The bytownite composition ($\text{An}_{72.7\text{--}70.3}\text{Ab}_{25.8\text{--}28.1}\text{Or}_{1.5\text{--}1.6}$) is predominantly found in crystal cores. Labradorite can occur from crystal cores to rims, while andesine ($\text{An}_{48.0\text{--}35.2}\text{Ab}_{47.4\text{--}57.0}\text{Or}_{4.6\text{--}7.8}$) forms intermediate areas (or rims) on crystals. The edges of crystals frequently exhibit overgrowths of quartz and sanidine ($\text{Or}_{90.7\text{--}42.0}\text{Ab}_{6.2\text{--}48.8}\text{An}_{3.1\text{--}9.2}$). Anorthoclase ($\text{Or}_{21.3}\text{Ab}_{59.4}\text{An}_{19.3}$) is rare but can also be observed in these alkalic shells ([Fig. 4D](#)). The degree of alteration of plagioclase ranges from low to moderate. Common secondary phases include albite, sericite, and clay minerals.

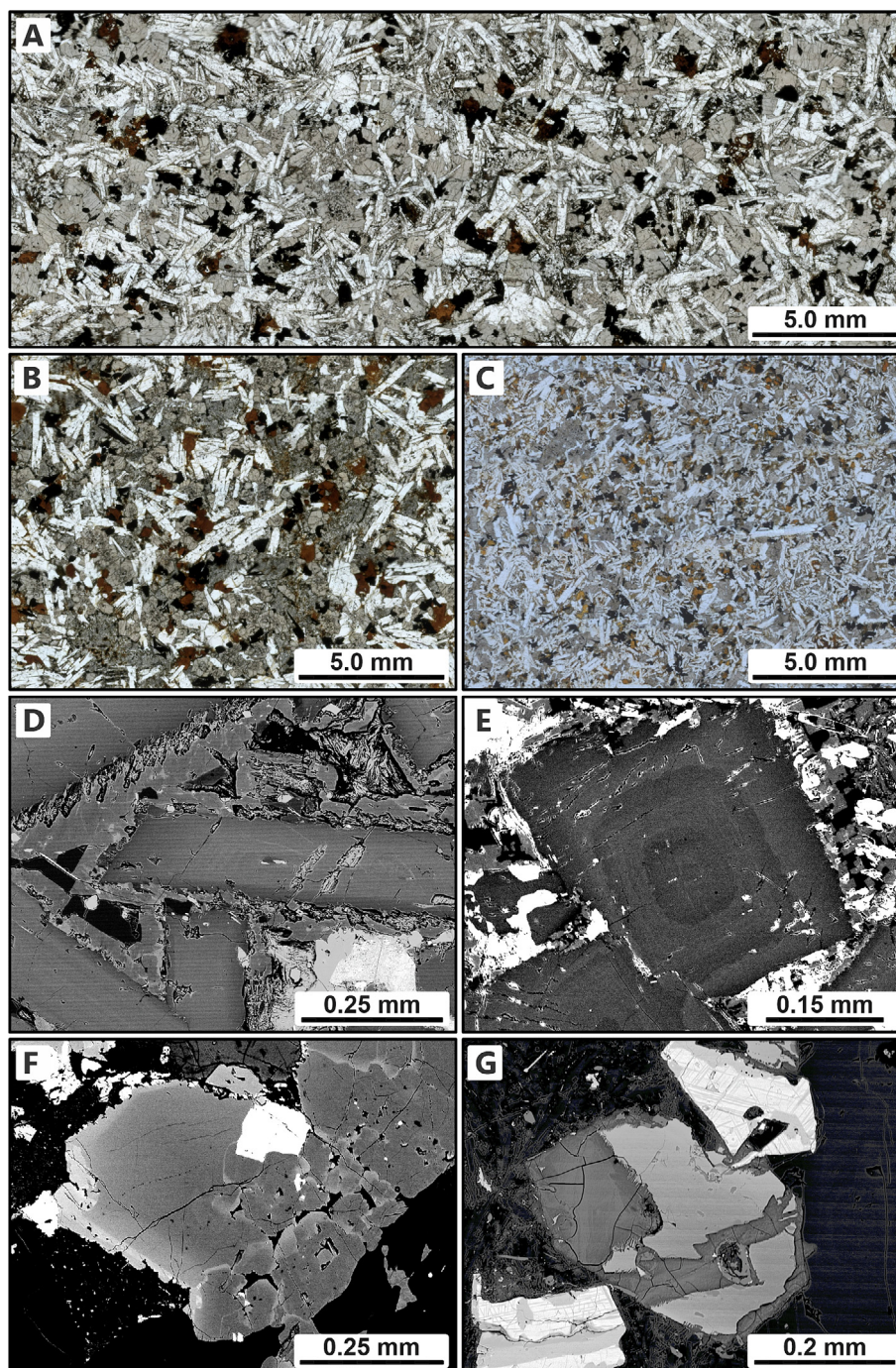


Fig. 4. Petrographic and textural features of the Transminas (Az-125, Az-170) and Pirenópolis dikes. (A) Typical microgabbroic texture of dikes, exhibiting fine to medium grains composed of plagioclase, pyroxene, Fe-Ti oxides, olivine, and interstitial glass. This texture is common across the swarm, with a trend toward finer grain size inland. Examples include (A) from the Az-170 swarm, (B) the southeastern portion of Az-125, and (C) the northwestern part of Az-125. (D) Feldspar crystals display either simple normal zoning or (E) complex zoning patterns, typically surrounded by alkali feldspar coronas. (F) Zoned clinopyroxene (augite) with a typical normal zoning pattern. (G) Partially altered olivine crystal in the Az-125 swarm.

Clinopyroxene is euhedral to anhedral (Fig. 4F), appearing as prismatic or tabular sections of purplish-brown crystals, reflecting the high Ti and Fe content. The global composition varies from augite to ferroaugite ($\text{En}_{39.3-9.6}\text{Wo}_{44.0-39.1}\text{Fe}_{16.7-51.3}$; Fig. 5; Supplementary Data Table S3). Ferroaugite is predominantly in the crystal rims ($\text{En}_{15.4-9.6}\text{Wo}_{39.6-39.1}\text{Fe}_{45.1-51.3}$), while diopside ($\text{En}_{36.6-35.0}\text{Wo}_{46.1-47.0}\text{Fe}_{17.2-18.0}$) is rare and found only in the crystal cores. Augite has normal zoning. The TiO_2 contents in the cores or intermediate positions of crystals are consistently > 0.7 wt.%, with an average of 1.0 wt.%. However, the augite rims have

$\text{TiO}_2 < 0.7$ wt.% (0.71–0.38 wt.%) with an average of 0.5 wt.%. Sieve textures are sometimes/often present in the grain cores, possibly related to the late destabilization of more primitive compositions. Minor alteration of clinopyroxene can occur, resulting in the formation of chlorite and Fe-Ti oxides.

Olivine is only found in the Transminas dikes. Olivine exhibits euhedral to subhedral morphology and often shows signs of fracturing and partial to total alteration (Fig. 4G). In the fresh relicts ($\text{Fo}_{45.6-26.0}$), the cores tend to be richer in the forsterite component than rims (Supplementary Data Table S4). Positive correlations

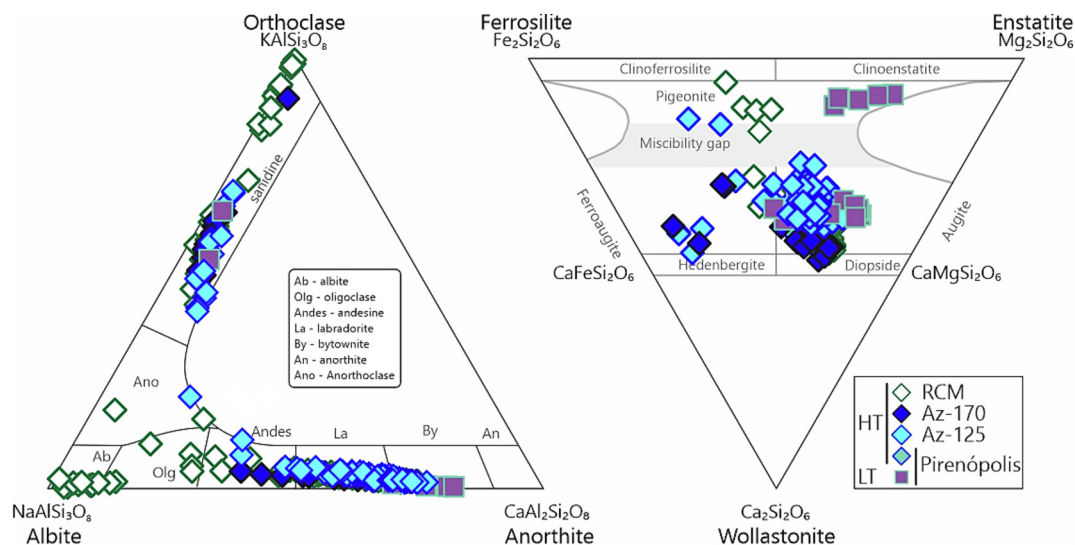


Fig. 5. Mineral chemistry data analyses for the Transminas (Az-125, Az-170) and Pirenópolis dikes: Feldspars and Pyroxenes. Magma groups are divided as HT (high-Ti tholeiites) and LT (low-Ti tholeiites). The dike swarms represented are Az-125, Az-170, and Pirenópolis. The HT of the Rio Ceará-Mirim dikes (RCM) are plotted for comparison (Macêdo Filho and Hollanda, 2022).

exist between MgO and Ni-Cr, while negative correlations between MgO and FeO (and MnO) are observed, supporting normal zoning. The common alteration phases are brownish/yellow to reddish-brown pseudomorphs. This mineral has a higher concentration of SiO₂ (48.8–44.6 wt.%) and Al₂O₃ (4.1–3.2 wt.%) compared to fresh olivine crystals, but lower levels of FeO_T (32.8–26.5 wt.%) and MgO (8.0–3.5 wt.%). The EMPA closure analysis yields a range of 87.8–86.3 wt.% ($n = 4$). Thus, optical characteristics and mineral chemistry support ferrous iddingsite as the main secondary phase of olivine. Additionally, there are minor amounts of greenish/gray minerals (chlorite, serpentine) and Fe-Mn oxides.

Opaque minerals are subhedral to euhedral (cubic, octahedral, and acicular). The examination of textural patterns under reflected light in a petrographic microscope, along with EMPA and EDS analysis, confirms the predominance of Fe-Ti oxides (titanomagnetite, ilmenite). Sulfides (pyrite, chalcocite, chalcopyrite) form late crystallizing phases. Some of these Fe-Ti oxides also form secondary phases from the destabilization of clinopyroxene. The EMPA results in titanomagnetite indicate Fe₂O₃^T contents from 89.4 wt.% to 80.2 wt.%, and TiO₂ from 20.3 wt.% to 10.6 wt.%, followed by minor amounts of Al₂O₃ from 2.6 wt.% to 1.5 wt.%. The other measured elements, Ca, Cr, K, Mg, Mn, Na, Nb, Ni, Si, and Zn, sum to less than 1.5 wt.% on average (Supplementary Data Table S5). Magnetite rich in Mn (FeO = 49.6 wt.%; MnO = 14.9 wt.%) is identified as a late crystallizing phase.

Apatite crystals in thin sections typically exhibit euhedral to subhedral morphology, displaying hexagonal shapes when observed in basal sections. They can also take on acicular habits, forming needle-like structures that appear as swarms within the rock texture when viewed in the elongation section. Zircon is not commonly observed in thin sections. When present, it tends to occur in intercrystalline pockets that are enriched in evolved melts and rich in quartz, alkali feldspar, apatite, zirconolite, and other late crystallizing phases. In addition, baddeleyite is observed, particularly included in pyroxene borders or melt pockets.

4.3. Unspiked K–Ar geochronology

We conducted dating at a total of 25 samples, performing triplicate analyses at each site (Supplementary Data Table S6). The resulting validated dates were then plotted on a binary diagram comparing dates versus the percentage of atmospheric Ar, facilitat-

ing the generation of a trend line. The approximate date is defined by the equation: $y = ax + b$, where y is the date, x represents the amount of atmospheric argon (contaminant), and b is the intercept. The best approximation of age for each dike swarm is when the sample has no atmospheric argon; thus, the value of intercept b when x is zero (Oliveira et al., 2021).

For the Az-170 swarm, the regression yielded a trend line defined by $y = -540.63x + 136.3$. The Az-125 swarm, in turn, produced a trend line of $y = -75.744x + 130.1$, while the high-Ti dikes from Pirenópolis yielded a trend line $y = -225.22x + 146.3$ (Fig. 6A). As observed, K–Ar estimated dates are 146.3 Ma, 136.3 Ma, and 130.1 Ma for the Pirenópolis, Az-170, and Az-125 swarms, respectively.

The application of this technique was fundamental in excluding potential Precambrian dikes from the analysis. For instance, at the southeastern termination of the Az-125 swarm, the Transminas dikes overlap with the Proterozoic Formiga and Pará de Minas dikes (Chaves and Correia Neves, 2005). In this sector, the exotic chemical composition and K–Ar dates of Sample DMG15 ranging from 727 Ma to 462 Ma (Supplementary Data Table S6) rule out any possible affiliation with the Paraná Province ($\sim 130 \pm 5$ Ma; Gomes and Vasconcelos, 2021). According to Caxito et al. (2020), this dike belongs to the Formiga swarm, dated at 897 ± 20 Ma (U–Pb). Since the sample shows extensive secondary alteration, it yielded younger ages with the K–Ar method.

The two samples of low-Ti dikes in the Pirenópolis swarm were not enough to generate a reliable trendline. The data from low-Ti dikes exhibited major heterogeneity, resulting in ages ranging from the Upper Triassic to the Middle Jurassic (229 Ma, 213 Ma, 196 Ma, 162 Ma) and two Paleozoic dates (345 Ma, 301 Ma). The Triassic–Jurassic dates yield a linear equation of $y = -12.039x + 202.3$ (Fig. 6B), consistent with the timing of the CAMP.

4.4. ⁴⁰Ar/³⁹Ar geochronology

To obtain more precise and accurate geochronological results, we chose to date samples from the Transminas and Pirenópolis dike swarms using the ⁴⁰Ar/³⁹Ar method (Fig. 7). Samples DMG01 and DMG14 correspond to Az-170 and Az-125 of the Transminas dikes, respectively, whereas samples DGO01 and DGO02 are from low-Ti dikes of the Pirenópolis Swarm. All samples

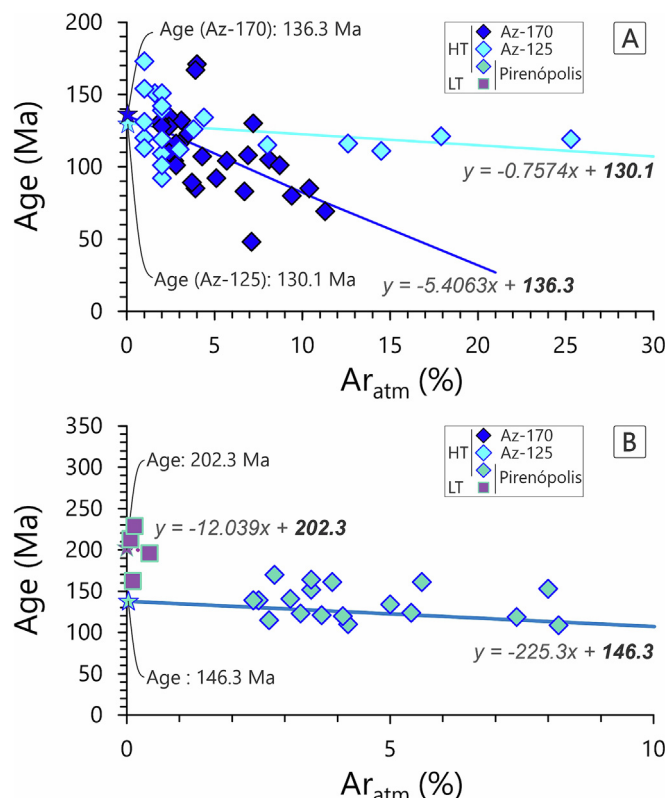


Fig. 6. Binary plots of Age (Ma) against % Atmospheric Argon ($Ar_{atm}\%$) from unspiked K-Ar analysis for the geochronology of the Transminas (Az-125, Az-170) and Pirenópolis dike swarms. (A) Az-125 and Az-170 from the Transminas dike dataset. (B) Pirenópolis dike dataset. The stars on the Y-axis represent the estimated ages based on the set of analyses for each dike swarm.

and standards were step-heated using a continuous-wave laser operated at progressively increasing power.

For sample DMG01, heating was carried out in successive power increments of 0.2 W, 0.3 W, and 0.5 W, followed by 1 W increases up to 8 W to achieve complete degassing. A total of twenty-two increments were analyzed. Atmospheric argon dominated the first, and to a lesser extent, the second increment. Seven consecutive intermediate- to high-temperature steps (I–O) yielded reproducible apparent ages within a 95% confidence level, defining a plateau age of 131.49 ± 1.90 Ma (1.49% uncertainty; MSWD = 2.1) corresponding to 51% of the released ^{39}Ar . The inverse isochron constructed to the plateau-defining steps yields an initial $^{40}\text{Ar}/^{36}\text{Ar}$ of 433 and a poorly constrained intercept age of 125.88 ± 4.33 Ma (Fig. 7). The corresponding K/Ca ratio is 0.29.

Sample DMG14 was analyzed in twenty steps following a similar heating protocol. Atmospheric argon was largely removed in the earliest increments, after which the steps showed internal consistency. From intermediate temperatures upward, the spectrum developed a discrete saddle-shaped pattern. The 95%-coherent steps define a plateau age of 133.08 ± 1.98 Ma (58% ^{39}Ar released; MSWD = 2.9), with a corresponding inverse isochron age of 130.83 ± 3.01 Ma and an initial $^{40}\text{Ar}/^{36}\text{Ar}$ intercept of 326 (Fig. 7).

The extraction protocol for the sample DGO01 began with fine increments from 0.1 to 1 W in 0.1 W steps, followed by intermediate increases of 0.2 W up to 2 W, and finally by higher-power increments of 0.3–0.5 W until reaching 8 W to ensure complete degassing. Atmospheric ^{40}Ar was efficiently removed within the first two increments, after which $^{36}\text{Ar}/^{40}\text{Ar}$ fell to <0.002 . A weighted-mean age of 196.39 ± 2.91 Ma (1.48% uncertainty; MSWD = 1.5) is defined by five consecutive high-temperature steps

(Q–U), plus one additional higher-temperature increment (W), together representing 57% of the cumulative ^{39}Ar released. The K/Ca ratio of the steps used for the age calculation is 0.04, and the corresponding inverse-isochron age is 196.14 ± 2.97 Ma, with a $^{40}\text{Ar}/^{36}\text{Ar}$ intercept of 302.2 ± 5.8 (Fig. 7).

Sample DGO02 was analyzed twice. In the first experiment, argon was extracted through progressively increasing heating increments like those applied to sample DGO01, beginning at low power and advancing to high-power steps, reaching 8 W to ensure complete degassing. This run produced a complex and disturbed spectrum with ^{40}Ar loss at lower temperatures, followed by a “half saddle” pattern indicative of excess ^{40}Ar in the analyzed fragments. A second rapid re-analysis using only thirteen heating increments reproduced the overall release behavior. Combined, the two experiments yield total-fusion ages of 205.5 Ma and 210.7 Ma (Fig. 7).

4.5. Major and trace elements

The high-Ti tholeiites exhibit evolutionary patterns characterized by a decrease in TiO_2 , CaO , and Fe_2O_3 , alongside an increase in K_2O as the MgO content declines (Fig. 8). The Al_2O_3 and Na_2O contents remain relatively constant despite the decrease in MgO . Positive correlations between MgO and CaO indicate the fractionation of plagioclase and augite, while decreases in TiO_2 and Fe_2O_3 reflect the progressive removal of Fe-Ti oxides (and to a lesser extent, augite). The increase in K_2O is consistent with the late crystallization of alkali feldspar. Samples from the Transminas dikes, which have lower SiO_2 contents, also contain olivine, while the Pirenópolis dikes, in contrast, are olivine-free tholeiites. The Formiga dike sample (DMG15) differs from the others by being more evolved, presenting higher SiO_2 (56.7 wt.%) and Al_2O_3 (14.4 wt.%), with low CaO (4.8 wt.%) and MgO (2.7 wt.%), and TiO_2 at 2 wt.% (Supplementary Data Table S1).

The high-Ti tholeiites have consistent characteristics in primitive mantle normalized multi-element diagrams (Fig. 9). They are characterized by elevated levels of LILEs (Rb, Ba, Th, U), which can reach a hundred times higher than the primitive mantle (Sun and McDonough, 1989), and a negative Nb-Ta anomaly. Positive Pb and Ti anomalies are observed in high-Ti types. The REE chondrite-normalized patterns (Boynton, 1984) display moderate fractionation. High-Ti tholeiites ($\Sigma\text{REE} = 188\text{--}275$ ppm) exhibit LREE values approximately a hundred times greater than chondrite, while HREE is elevated by about 20 times, therefore yielding La/Yb_{cn} values from 7.5 to 10.4. A steep slope between the middle and heavy REE likely indicates residual garnet during melting. Sample DMG15, linked to the Formiga swarm (see Section 4.7), differs from the others by being more enriched in LILE, Nd, Sm, and Eu, accompanied by negative Ti anomalies (Fig. 8).

The low-Ti tholeiites are poor in Fe_2O_3 , Na_2O , K_2O , and P_2O_5 , and richer in MgO (>6.5 wt.%), CaO , and Al_2O_3 (Fig. 8). The trace element pattern of low-Ti tholeiites closely resembles that of the high-Ti types, but with lower HFSE (Nb, Ta) and LILEs (Fig. 9). The primary distinction lies in the degree of enrichment, which can achieve thirty times higher than primitive mantle values (Sun and McDonough, 1989). The low-Ti tholeiites ($\Sigma\text{REE} = 71.5\text{--}74.2$ ppm) show less than thirty-fold increases compared to chondritic values (Boynton, 1984), but their HREE abundance resembles that observed in the high-Ti suite. Thus, La/Yb_{cn} values range around 3.1–3.0. The Formiga dike sample (DMG15) displays a relative enrichment in HREEs compared to the other dikes.

The Nb/Ta ratios vary from 17.4 to 11.6 (averaging 15.5), remaining constant concerning MgO content variance, while the Zr/Hf ratios range from 45.2 to 37.2 (averaging 41.2). Lower values are observed for the low-Ti rocks, which have Nb/Ta from 14.9 to 14.8, and Zr/Hf from 38.9 to 37.8. In the Zr/Hf versus Nb/Ta diagram (Pfänder et al., 2007), all studied suites plot above the over-

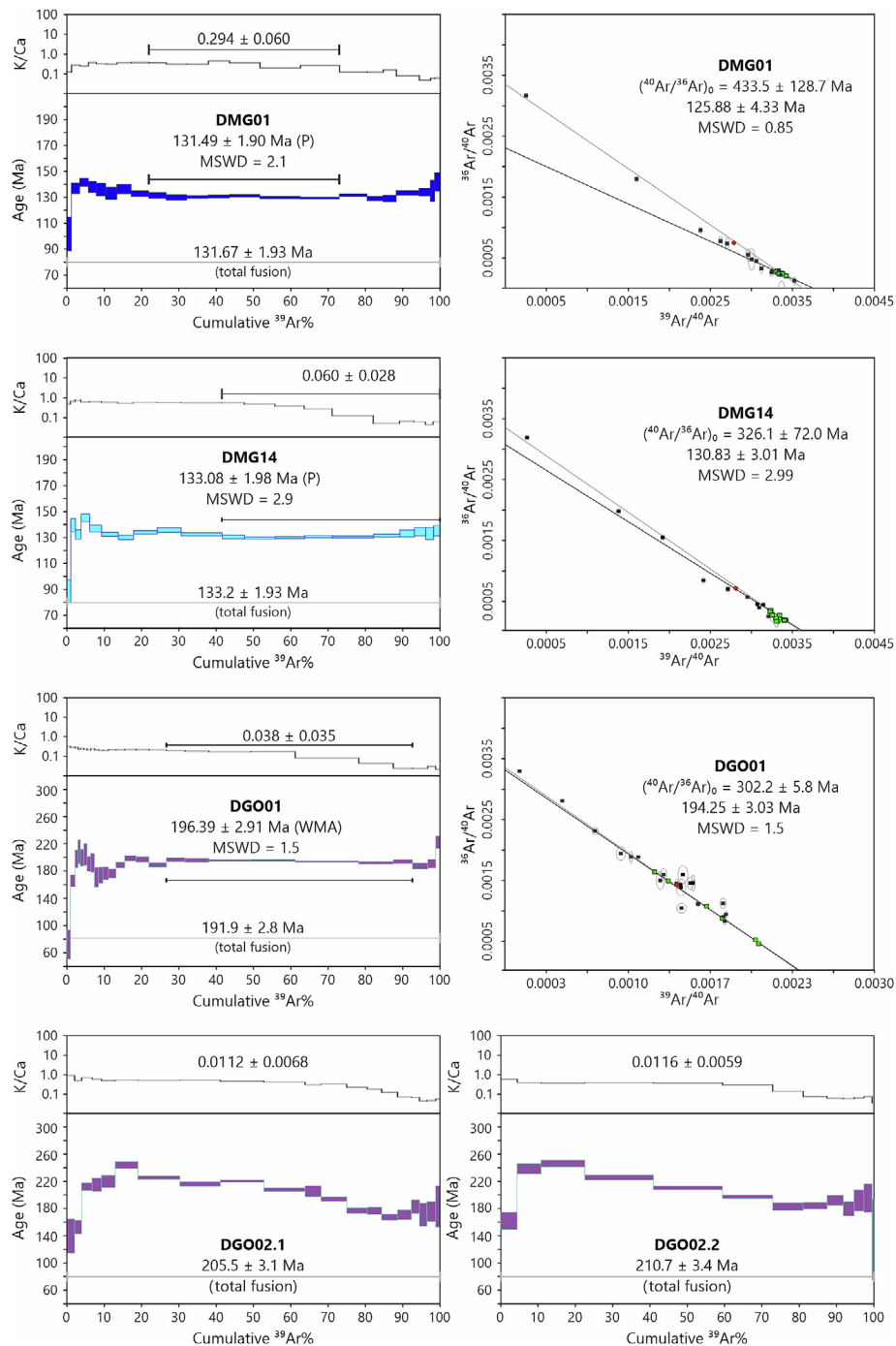


Fig. 7. Apparent $^{40}\text{Ar}/^{39}\text{Ar}$ age spectra and K/Ca compositions versus cumulative ^{39}Ar (%) released (left) for samples of the Transminas/Az-170 (DMG01), Transminas/Az-125 (DMG14), and Pirenópolis dikes (DGO01, DGO02). Green squares represent the steps included in the isochron calculation, whereas black squares indicate steps that were excluded. The red circle marks the Total Fusion (TF) age. Inverse isochrons for the first three samples are also shown (right). Full name of abbreviations: P, plateau age; WMA, weighted mean age.

lapping zone of OIBs, MORBs, and continental basalts (not shown). Analyzing the entire spectrum of trace elements, the samples are fully situated within the compositional envelope of high-Ti tholeiites from the PEMP and EQUAMP (Macêdo Filho et al., 2023a), exhibiting incompatible element concentrations comparable to those found in OIB melts (Sun and McDonough, 1989); notably, the sole distinction lies in the presence of Nb-Ta negative anomalies and a marginally elevated concentration of LILEs relative to typical OIBs. The low-Ti samples, in turn, are poorer in incompatible elements, analogous to low-Ti flood basalts of the Mesozoic

LIPs in West Gondwana (e.g., Peate et al., 1992; Marzoli, 1999; Marzoli et al., 2018; Pearce et al., 2021; Macêdo Filho et al., 2023a,b).

4.6. Radiogenic isotopes (Sr, Nd, Pb)

The isotopic compositions of Sr, Nd, and Pb were analyzed for twenty-four samples from the Az-170, Az-125, and Pirenópolis dikes (Supplementary Data Table S1). In addition, we included five samples from previous publications (Rosset et al., 2007; Marques

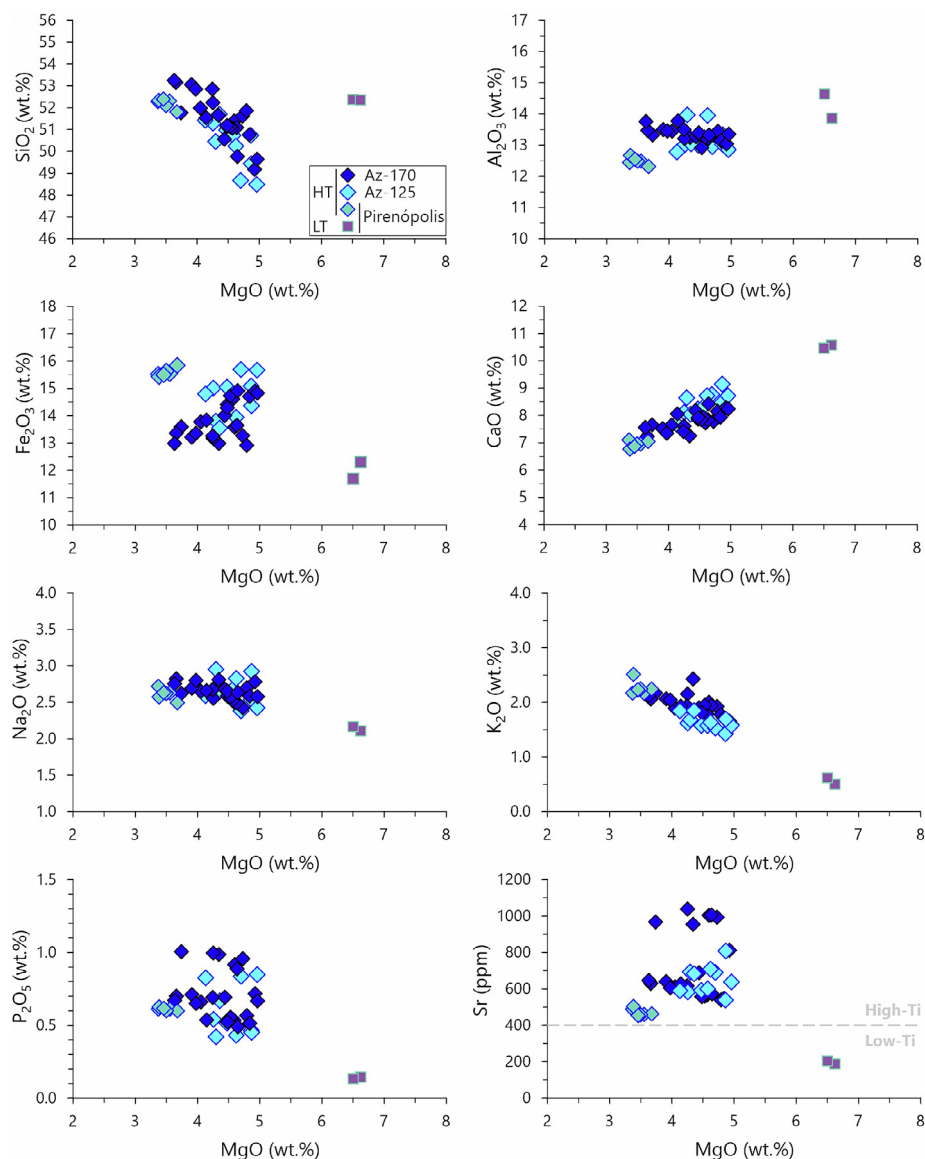


Fig. 8. Binary plots showing MgO versus major oxides and Sr for the Transminas and Pirenópolis dike swarms. Magma groups are divided as HT (high-Ti tholeiites) and LT (low-Ti tholeiites).

et al., 2016) in our new isotopic dataset to provide a more comprehensive characterization of the studied dike swarms. All discussed ratios are recalculated to 130 Ma. This age was chosen considering the peak of magmatic activity in the SAMP (Macêdo Filho et al., 2023a).

The $^{87}\text{Sr}/^{86}\text{Sr}_{(i)}$ and $^{143}\text{Nd}/^{144}\text{Nd}_{(i)}$ in the high-Ti dikes of the Az-170 swarm range from 0.707321 to 0.706578, and from 0.512423 to 0.512423 ($\epsilon_{\text{Nd}} = -0.93$ to -6.05 ; av. -3.4 ; Fig. 10A, B), respectively. The T_{DM} ages of these dikes range from 1.4 Ga to 0.9 Ga (av. 1.1 Ga). The samples DMG04 and DMG09 presented heterogeneous behavior with $^{87}\text{Sr}/^{86}\text{Sr}_{(i)} = 0.708292$ – 0.708208 and $^{143}\text{Nd}/^{144}\text{Nd}_{(i)} = 0.511788$ – 0.511639 ($\epsilon_{\text{Nd}} = -13.3$ to -16.2 ; $T_{\text{DM}} = 2.3$ – 2.0 Ga) that might indicate greater degrees of assimilation of old crustal lithologies.

The Az-125 dikes show $^{87}\text{Sr}/^{86}\text{Sr}_{(i)}$ values between 0.70604 and 0.70549, and $^{143}\text{Nd}/^{144}\text{Nd}_{(i)}$ from 0.512366 to 0.512076 ($\epsilon_{\text{Nd}} = -2.0$ to -3.3 ; av. -2.5 ; Fig. 10A, B). The sample DMG12 presented less radiogenic Nd composition, with $^{143}\text{Nd}/^{144}\text{Nd}_{(i)} = 0.512075$ ($\epsilon_{\text{Nd}} = -7.7$). In this case, the T_{DM} ages are between 1.6 Ga and 1.0 Ga (av. 1.2 Ga).

The high-Ti tholeiites of the Pirenópolis dikes have $^{87}\text{Sr}/^{86}\text{Sr}_{(i)}$ and $^{143}\text{Nd}/^{144}\text{Nd}_{(i)}$ ratios ranging from 0.71157 to 0.70675, and 0.51275 to 0.512221 ($\epsilon_{\text{Nd}} = -1.8$ to -4.8 ; av. -2.8). For these dikes, the T_{DM} ages range from 1.25 Ga to 1.0 Ga (av. 1.1 Ga). The low-Ti dikes of the Pirenópolis area have $^{87}\text{Sr}/^{86}\text{Sr}_{(i)}$ ranging from 0.70651 to 0.70615 (Fig. 10A, B), and $^{143}\text{Nd}/^{144}\text{Nd}_{(i)}$ from 0.512524 to 0.512483 ($\epsilon_{\text{Nd}} = +0.24$ to $+1.03$). For low-Ti dikes, the T_{DM} ages range from 1.03 Ga to 0.77 Ga. For lead isotopes (Fig. 10C, D), all high-Ti dikes plot above the Northern Hemisphere Reference Line (Hart, 1984) and to the right of the geochron. The samples exhibit a narrow range in Pb isotope values, with $^{206}\text{Pb}/^{204}\text{Pb}_{(m)}$ ranging from 18.40 to 18.12, $^{207}\text{Pb}/^{204}\text{Pb}_{(m)}$ from 15.64 to 15.54, and $^{208}\text{Pb}/^{204}\text{Pb}_{(m)}$ from 38.44 to 38.76. The two samples from the low-Ti dikes of the Pirenópolis are the most radiogenic in Pb.

The high-Ti samples plot within an EMI-type field (Sun and McDonough, 1989; Stracke et al., 2005), similar to the high-Ti tholeiites of the PEMP, such as Florianópolis dikes, Bero Volcanic complex (Angola), Urubici/Khumib- and Pitanga-type, as well as high-Ti tholeiites from the EQUAMP intrusions (Macêdo Filho et al., 2023a). The two low-Ti samples, in turn, plot exactly within

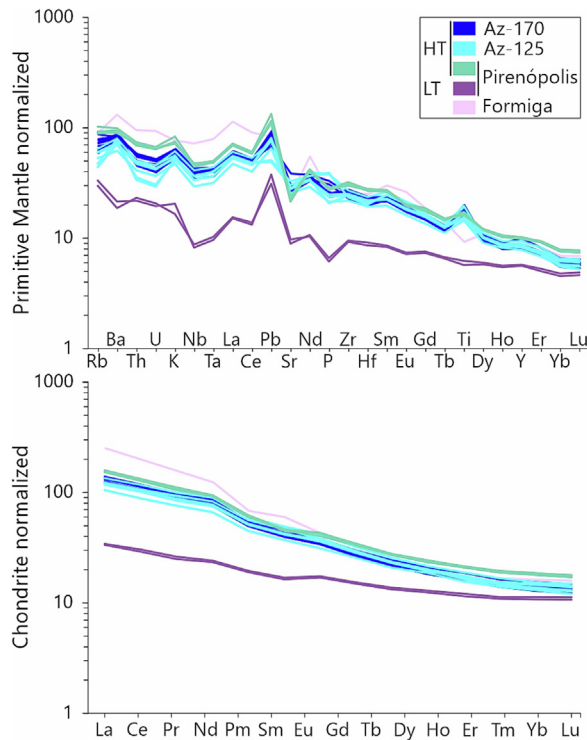


Fig. 9. Multi-element and Rare Earth Element (REE) diagrams for the Transminas (Az-125, Az-170), Pirenópolis, and Formiga dike swarms. Magma groups are divided as HT (high-Ti tholeiites) and LT (low-Ti tholeiites). Multi-element data are normalized to Primitive Mantle values from Sun and McDonough (1989), while REE patterns are normalized to Chondrite values from Boynton (1984).

the low-Ti field of the Triassic dikes of the Borborema Province and the Mosquito/CAMP magmatism in the Parnaíba Basin (Macêdo Filho and Hollanda, 2022; Macêdo Filho et al., 2023b).

5. Discussion

5.1. Petrogenesis

The low-Ti magma types of the PEMP and EQUAMP (Fig. 11A) show pronounced negative Nb–Ta anomalies and positive Pb spikes (arc-like basalts). The high-Ti tholeiites, in turn, are generally more enriched in trace elements overall, but likewise display negative Nb–Ta anomalies (Fig. 11B). This observation may indicate a minor contribution from the continental crust to the high-Ti melts (De Min et al., 2018; Zhou et al., 2020; Macêdo Filho et al., 2026a). Similar interpretations arise from the LIP printing diagram (Pearce et al., 2021), where the high-Ti tholeiites are confined within the EM–OIB field (Fig. 11C), while the low-Ti samples plot within the subduction-modified lithospheric mantle (SZLM) domain. According to Pearce et al. (2021), the EM–OIB field includes samples with high TiO_2/Yb (and La/Yb), which serve as proxies for residual garnet, suggesting deeper (and/or lower-degree of) partial melting. In contrast, the SZLM types are characterized by high Th/Nb ratios, indicative of crustal input, primarily associated with subduction processes and/or crustal assimilation.

The low-Ti tholeiites of the Pirenópolis dikes represent subvolcanic equivalents of basaltic andesites, comparable to the prevalent magma type of the CAMP (Fig. 11A). These rocks exhibit geochemical characteristics like those of continental basalts derived from lithospheric sources. According to Marzoli et al. (2018), the predominant volumetric component of CAMP basalts is the depleted upper mantle, with a small amount of crustal contamination. In contrast, high-Ti CAMP basalts in South America exhibit depletion in incompatible elements and a trace element pattern comparable to enriched MORBs. These rocks show no significant evidence of crustal assimilation, as indicated by Sr, Nd, and Pb isotopic constraints. Therefore, we agree that a possible explanation for the petrogenesis of low-Ti tholeiites is that (E) MORB melts may have assimilated lithospheric materials, giving

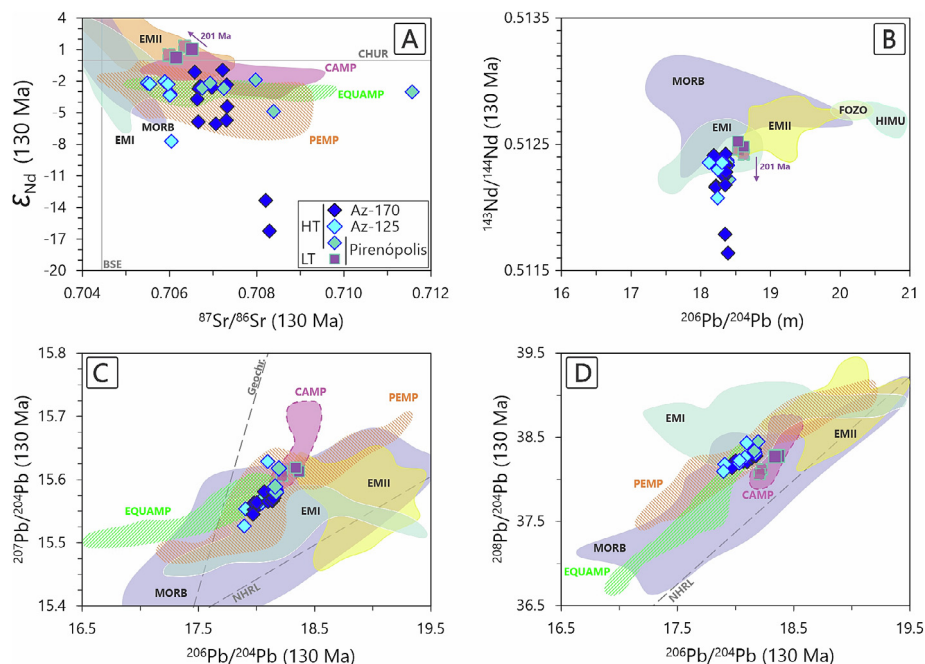


Fig. 10. Isotopic compositions of the Transminas (Az-1250, Az-170) and Pirenópolis dike swarms. The fields representing PEMP (Paraná–Etendeka Magmatic Province), EQUAMP (Equatorial Atlantic Magmatic Province), and CAMP (Central Atlantic Magmatic Province) datasets were compiled from Florisbal et al. (2018), De Min et al. (2018), Marques et al. (2018), Macêdo Filho and Hollanda (2022), and Macêdo Filho et al. (2023a,b). All values are age-corrected to 130 Ma; however, CAMP rocks are also shown age-corrected to 201 Ma (dashed and transparent symbols). MORB (Mid-Ocean Ridge Basalt) and OIB (Ocean Island Basalt) endmember fields are based on Stracke et al. (2005). Full name of other endmembers: EMI, Enriched Mantle I; EMII, Enriched Mantle II; FOZO, Focal Zone; HIMU, High μ mantle ($\mu = {}^{238}\text{U}/{}^{204}\text{Pb}$).

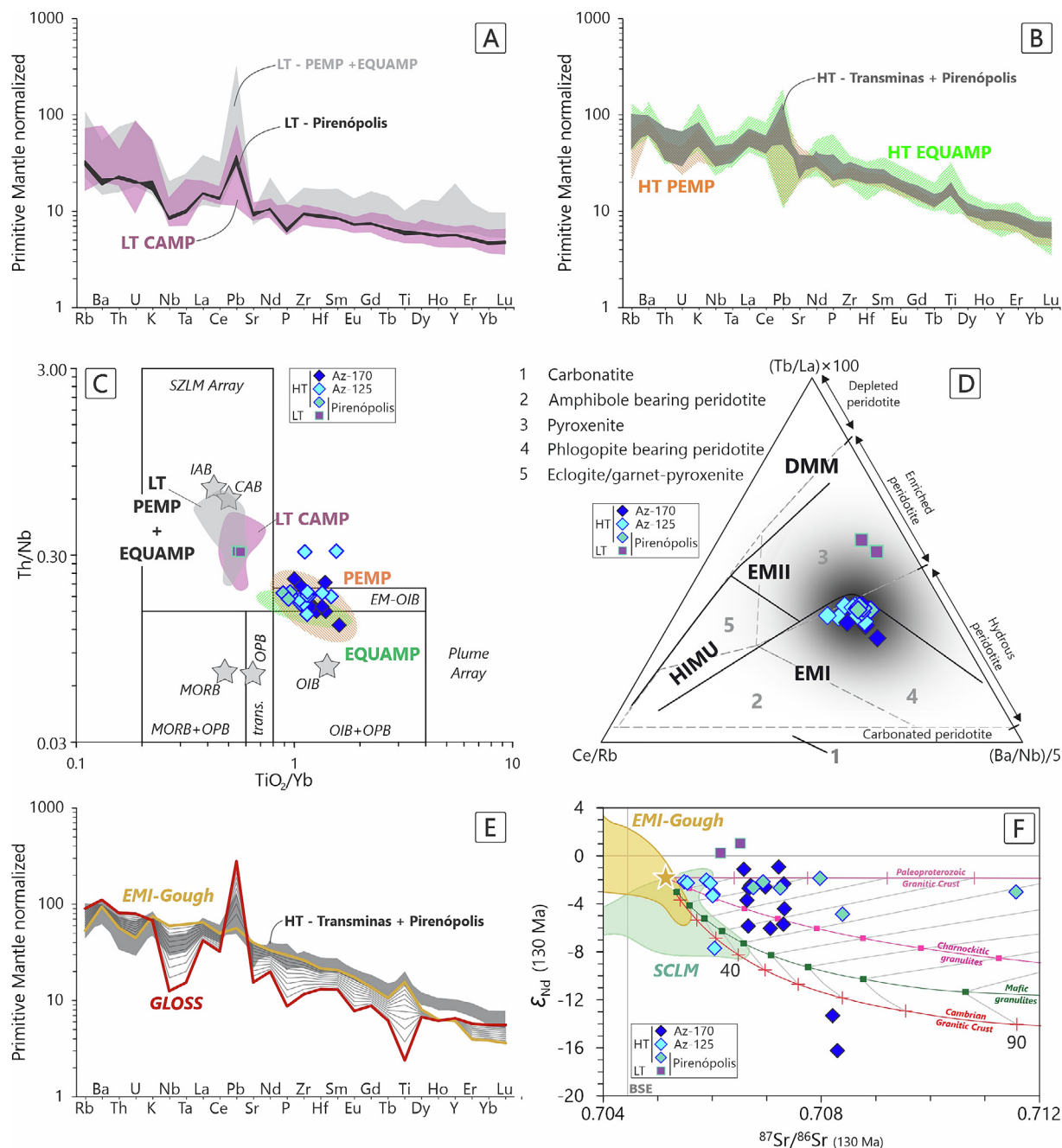


Fig. 11. The Transminas (Az-125, Az-170) and Pirenópolis dike swarms dataset plotted on petrogenetic diagrams. In Addition, we also plotted one sample from the Precambrian Formiga dikes. Comparison of trace element patterns of (A) low-Ti (LT) and (B) high-Ti (HT) tholeiites of the Transminas and Pirenópolis dikes with the PEMP, EQUAMP, and CAMP datasets (compiled from De Min et al., 2018; Macêdo Filho and Hollanda, 2022; Macêdo Filho et al., 2023a,b). (C) Two-proxy projection of Th/Nb versus TiO_2/Yb (Pearce et al., 2021). (D) Ternary diagrams for mantle endmembers and sources based on trace element ratios (Akbari et al., 2023). (E) Trace element mixing models showing the mixing of an EMI-Gough member (Hoernle et al., 2015) with crustal sources represented by the Global Subducted Sediment composition (GLOSS; Plank and Langmuir, 1998). Gray lines represent 10% of the mix between the two components. (F) Isotopic assimilation and fractional crystallization (AFC) models considering an EMI-Gough melt as the primary source contaminated by Proterozoic and Cambrian continental crust of the northern Mantiqueira Province. The SCLM (Subcontinental Lithospheric Mantle) field is also shown. The database was compiled from Oliveira (2002), Martins et al. (2004), Hoernle et al. (2015), and Araujo et al. (2019). All multi-element data are normalized to Primitive Mantle values from Sun and McDonough (1989). Full name of abbreviations: PEMP, Paraná-Etendeka Magmatic Province; EQUAMP, Equatorial Atlantic Magmatic Province; CAMP, Central Atlantic Magmatic Province; SZLM, Subduction Zone Lithospheric Mantle; MORB, Mid-Ocean Ridge Basalt; OPB, Oceanic Plateau Basalt; IAB, Island Arc Basalt; CAB, Continental Arc Basalt; EM, Enriched Mantle; OIB, Ocean Island Basalt; EMI, Enriched Mantle I; EMII, Enriched Mantle II; HIMU, High μ mantle ($\mu = ^{238}U/^{204}Pb$).

rise to the prevalent low-Ti tholeiites of the CAMP (Macêdo Filho et al., 2023b).

There is some isotopic resemblance between high-Ti types of the SAMP and basalts from Gough Island (Hoernle et al., 2015). The similarities are not only in terms of whole rock compositions (Fig. 11C, D, E); Hf isotope data in zircons from high-Ti basalts of

the PEMP (Hartmann et al., 2019) also show ϵ_{Hf} values analogous to those of basalts of the Gough Island (Salters et al., 2011). Thus, assuming an EMI-Gough-like composition for the parental melt (e.g., Ewart et al., 2004; Hoernle et al., 2015; Zhou et al., 2020), only minor contributions from crustal material – here approximated by the GLOSS (Global Subducting Sediment) composition (Plank and

Langmuir, 1998) – are required to account for the trace element abundances of the high-Ti tholeiites (Fig. 11E). Conversely, the assimilation–fractional crystallization (AFC) modeling performed using the same parental melt and representative isotopic compositions of the Proterozoic–Cambrian crust of the Mantiqueira Province indicates that most samples experienced low degree of crustal contamination (Fig. 11E), with assimilation values on the order of 5%–10% (r -value at 0.25; Tegner et al., 2005). This proportion may be even smaller if we consider that lateral scatter in the Sr decoupled from the Nd isotope ratios could be attributed to some post-magmatic alteration (Macêdo Filho and Hollanda, 2022). As observed, the magmas require minimal contributions from the SCLM to account for their predominant isotopic compositions. However, since there are no strong isotopic gradients between the EMI–Gough plume and EMI–flavored SCLM of West Gondwana, accurately determining the exact proportion of each source involved remains challenging.

Another possibility is the potential involvement of a HIMU-type component, given that the northern Paraná was positioned over the Saint Helena plume during the Early Cretaceous (Ernesto et al., 2002; Scotese, 2016). In this scenario, HIMU sources would mix with approximately 20% EM melts derived from the SCLM to produce the high-Ti tholeiites (Macêdo Filho and Hollanda, 2022). It is also possible that following lithospheric thinning, a significant amount of this enriched asthenosphere (re)equilibrated within the SCLM domain, driven by thermal subsidence associated with continental drift. Much of this ‘brand-new’ SCLM could have subsequently served as a source for the Late Cretaceous/Cenozoic alkaline magmatism that geographically coincides with the PEMP. The alkaline/carbonatitic magmatism in southeastern South America exhibits compositions intermediate between the HIMU and EMI components (Gomes, 2025).

5.2. Considerations about geochronology

The K–Ar technique constituted an initial step for rapid geochronological screening. Owing to its straightforward application, it was employed across all samples characterized in this study. These results subsequently guided the selection of the most suitable samples for $^{40}\text{Ar}/^{39}\text{Ar}$ dating. As anticipated, samples that exhibited superior performance in the K–Ar analyses also produced more reliable $^{40}\text{Ar}/^{39}\text{Ar}$ ages, underscoring the effectiveness of this strategy for preliminary sample selection.

When considering the $^{40}\text{Ar}/^{39}\text{Ar}$ database available in the literature for mafic rocks from the CAMP and SAMP, two apparent peaks of magmatic activity emerge at 199.1 Ma and 134.5 Ma (Gomes and Vasconcelos, 2021; Macêdo Filho et al., 2023a,b). For CAMP, high-precision U–Pb zircon CA-ID-TIMS geochronology provided emplacement ages clustering around ~201.5 Ma (Davies et al., 2017; Oliveira et al., 2023). Existing data for silicic lavas of the SAMP suggest a younger peak of magmatic activity (Rocha et al., 2020; Sun et al., 2025) than that proposed by $^{40}\text{Ar}/^{39}\text{Ar}$ studies (Gomes and Vasconcelos, 2021). Both LIPs, therefore, reveal discrepancies among results obtained using U–Pb and $^{40}\text{Ar}/^{39}\text{Ar}$ methods. Processes such as radiogenic ^{40}Ar loss (e.g., through weathering, devitrification, or carbonate formation), incorporation of excess or extraneous ^{40}Ar (for example, through crustal contamination), partial resetting during low-grade thermal events, or the presence of mineral inclusions with differing ^{40}Ar retentivities can all shift $^{40}\text{Ar}/^{39}\text{Ar}$ ages toward geologically meaningless, apparent dates. Despite these limitations, the new $^{40}\text{Ar}/^{39}\text{Ar}$ results presented here, when integrated with K–Ar screening criteria and comprehensive geochemical characterization, reinforce the coexistence of CAMP and SAMP magmatic events in central and southeastern Brazil.

5.3. Dikes as feeder systems of flood basalts?

In the Paraná Province, the high-Ti–Fe tholeiites (Pitanga-type) form extensive flood basalts and manifest as sill complexes and dike swarms (Renne et al., 1996; Janasi et al., 2011; Rocha-Júnior et al., 2013; De Min et al., 2018; Moraes et al., 2018). The high-Ti–Sr (Urubici-type) is well-documented as flood basalts in southern Brazil, Florianópolis dikes (Fig. 12A, B), and northern Etendeka Province, where they are known as Khumib (Peate, 1997; Marsh et al., 2001; Florisbal et al., 2018; Marques et al., 2018). High-Ti–Sr tholeiites are reported in the Serra do Mar, Resende-Ilha Grande, and Transminas swarms (Rosset et al., 2007; Corval et al., 2008; Coelho and Chaves, 2016; Guedes et al., 2016; Marques et al., 2016; this work). Those magmas correlate to some degree with the Urubici-type (high Sr); however, they can be slightly richer in Fe_2O_3 , as observed in the Pitanga-type (Fig. 12A, B). Similar high-Ti–Sr–Fe tholeiites occur in the EQUAMP intrusions (Ngonge et al., 2016; Macêdo Filho and Hollanda, 2022; Macêdo Filho et al., 2023a,b,c).

As shown in Fig. 1, the Az-125 dike swarm is closely associated with substantial volumes of flood basalts in the northern flank of the Paraná Basin. The northern Paraná Basin also contains occurrences of high-Ti–Sr–Fe flood basalts (Janasi et al., 2011; Machado et al., 2007). This suggests that the Transminas dikes might have acted as feeder systems for tholeiitic volcanism in northern Paraná, much like the Florianópolis (Urubici and Pitanga types) and Ponta Grossa (Paranapanema, Pitanga, and Ribeira types) swarms fed flood basalts in the central Paraná Province (Piccirillo et al., 1990; Renne et al., 1996; Florisbal et al., 2018). There is a trend of grain-size reduction when comparing the microgabbros in the southeastern part of AZ-125 with those in the northwestern termination and the high-Ti Pirenópolis dikes. This suggests that the crustal level exposed in the more intracontinental sectors corre-

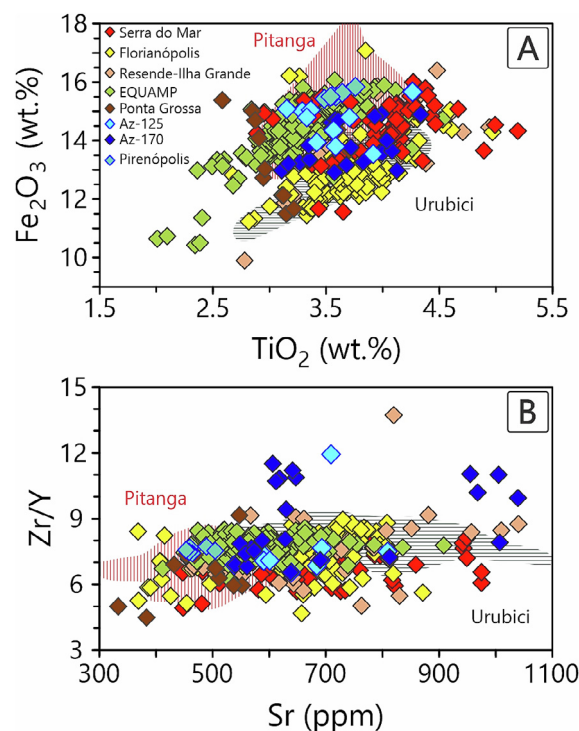


Fig. 12. Samples of high-Ti tholeiites from the PEMP (Paraná–Etendeka Magmatic Province) and EQUAMP (Equatorial Atlantic Magmatic Province) are plotted in (A) TiO_2 (wt.%) versus Fe_2O_3 (wt.%) and (B) Sr versus Zr/Y diagrams, which are used to discriminate high-Ti basalt types of the PEMP (Peate et al., 1992). The geochemical dataset sources: Renne et al. (1996); Corval et al. (2008); Guedes et al. (2016); Ngonge et al. (2016); Marques et al. (2016, 2018); Florisbal et al. (2018); Macêdo Filho and Hollanda (2022); Macêdo Filho et al. (2023b).

sponds to shallower sections, compared to the dikes located closer to the South Atlantic margin, which were more strongly exhumed during the post-breakup stage (Fonseca et al., 2022). This phenomenon is also observed in the EQUAMP (Macêdo Filho et al., 2023c).

Although extrusive rocks are absent in the EQUAMP – either due to their non-emplacement or subsequent denudation during the post-breakup period, driven by strong exhumation along the onshore equatorial margin of South America (Fellin et al., 2024; Macêdo Filho et al., 2025) – the Rio Ceará-Mirim dikes, which initially propagated continentward, likely acted as feeder conduits for mafic magmas emplaced in the Parnaíba Basin (Macêdo Filho et al., 2023c). Two zircon populations dated via CA-ID-TIMS – one corresponding to the age of the dikes (133.363 ± 0.029 Ma) and another slightly younger (133.071 ± 0.031 Ma) support the interpretation that Sardinha sills were fed by dikes, with the younger population possibly growing during the final crystallization of the sills (Oliveira et al., 2025).

In the CAMP, mafic dike swarms are intrusive in the Precambrian basement, while in the Paleozoic intracratonic basins, the magmas were emplaced mainly as sills and flood basalts. This is well exemplified by the NW-SE branch of mafic dikes in Amapá (South America) and Guinea/Liberia (Africa), which appear to form a single swarm that, upon reaching the Parnaíba Basin, fed flood basalts and sills (Heilbron et al., 2018; Macêdo Filho et al., 2023c). The NW-SE branch of the CAMP presents the oldest ages of the LIP, namely 201.614 ± 0.029 Ma in Guinea and 201.579 ± 0.057 Ma in the Borborema Province (Oliveira et al., 2023). Spreading rates may have been even higher when considering the NE-SW axis of the CAMP, as the basalts in Bolivia are essentially coeval with the intrusions in Guinea, although the total duration of CAMP activity lasted at least ~ 700 kyr (Oliveira et al., 2023). The availability of extensive continental areas of Pangea may have enabled the development of large magma plumbing systems, which in turn contributed to the broad geographic distribution of the CAMP.

When comparing the Early Cretaceous dike patterns in West Gondwana (Fig. 1A), a striking contrast becomes apparent: while large orthogonal swarms dominate in South America, the African side is mostly characterized by dikes oriented longitudinally along the South Atlantic coast (except for the HOD swarm). One possible explanation is the development of an asymmetric radial intrusive pattern from volcanic centers. Following the model of Ernst and Buchan (2001), the Transminas dikes were emplaced in a homogeneous domal uplift context, resulting in a fan-shaped pattern opening toward South America. In contrast, in the EQUAMP area, the domal uplift is associated with triple-junction rifting, producing a giant arcuate dike swarm and aborted set of rift basins (Cariri-Potiguar Rift System). Therefore, we envisage that as these extensive plumbing systems advanced into continental inland areas, they likely contributed to the emplacement of flood basalts and sills, whose remnants are currently preserved in the Paraná and Parnaíba basins. Furthermore, it is reasonable to consider that substantial portions of these flood basalts have been eroded over time, as major erosion events have shaped the onshore South Atlantic margin since the continental breakup (e.g., Fonseca et al., 2022). A significant portion of the massive flood basalt volumes, once weathered and eroded during the early stages of emplacement, may have played a key role in triggering the Weissert Event – a Valanginian global environmental perturbation predominantly characterized by climatic cooling (Oliveira et al., 2025).

5.4. Geodynamics

The trigger mechanisms of LIPs remain a topic of scientific debate, involving a spectrum of plume and non-plume models (Ernst, 2014). A variety of petrogenetic and geodynamic scenarios

have been proposed to explain the origin of the high-Ti tholeiites in the PEMP area, reflecting the complex interplay between mantle dynamics and lithospheric architecture. They include partial melting of: (1) delaminated SCLM recycled by mantle plumes (Peate et al., 1999); (2) asthenosphere enriched by ancient subduction processes (Rocha-Júnior et al., 2012); (3) enriched materials from the LLSVPs mobilized by mantle plumes (Hoernle et al., 2015); (4) SCLM triggered by mantle plume (Beccaluva et al., 2020); (5) SCLM modified by carbonatite metasomatism (De Min et al., 2018); (6) pyroxenites recycled by mantle plume activity (Zhou et al., 2020); (7) asthenosphere hybridized with recycled crustal components during the initial stages of subcontinental plume head development (Pearce et al., 2021). What is common to all these models is the requirement of enriched mantle sources to account for the genesis of high-Ti basalts.

As shown in Fig. 9, the Sr, Nd, and Pb isotopic ratios of the high-Ti types from PEMP and EQUAMP overlap with the EMI field, consistent with their trace element compositions (Fig. 11D). In this context, one conservative interpretation is that the mantle beneath southeastern (PEMP) and northeastern (EQUAMP) South America shared similarly enriched characteristics, giving rise to comparable high-Ti magmatism in these distant regions. Within the framework of a passive asthenospheric upwelling model, the sources of the high-Ti tholeiites would include the common asthenosphere ($\sim$ DMM) and the SCLM of West Gondwana (Macêdo Filho and Hollanda, 2022). On the other hand, in an active rifting model, the generation of high-Ti tholeiites would be governed by an EMI-flavored mantle plume, characterized by a significant thermo-mechanical budget.

According to Hassan et al. (2015), a localized thermal anomaly may have influenced the southern portion of the PEMP, which was later followed by the ascent of a more robust anomaly in the northern PEMP. The substantial volume of tholeiitic basalts, low-Ti (E-MORB) dikes, alkaline/carbonatite/kimberlite complexes, and picrites reinforces the Etendeka region as a key focal area for plume-related magmatism (e.g., Hoernle et al., 2015; Stronick et al., 2017; Beccaluva et al., 2020; Zhou et al., 2020; Macêdo Filho et al., 2026a). Moreover, a prominent mantle dome beneath the northern PEMP (Angola; Fig. 13) suggests that lithospheric excavation was induced by interaction with the Tristan-Gough plume (Celli et al., 2020). In its South American counterpart (Mantiqueira Province), a series of large-scale dike systems can be categorized as either radial (Transminas and Vitória dikes) or axial (Serra do Mar, Resende-Ilha Grande), in alignment with the hypothesis of magma center in the region (Fig. 13). This area is also characterized by the presence of E-MORBs (Almeida et al., 2021; Carvas et al., 2021), which may further support the existence of a depleted upper mantle that has been enriched by an OIB component (Macêdo Filho et al., 2023a, 2026a). In this context, we could also infer other lithospheric dome structures in the EQUAMP and Etendeka areas (Fig. 13), suggesting that the SAMP may have formed due to multiple magma centers and their stress interactions, as has been well documented in both terrestrial and Venusian LIPs (Ernst, 2014; Ben Marzoug et al., 2024; El Bilali and Ernst, 2024).

Another possibility is the source (and/or magma) migration across the South Atlantic Rift System axis. As a dynamic process, portions of the Tristan-Gough plume may have migrated laterally over long distances beneath thinned lithospheric domains, similar to what has been documented for other LIPs (e.g., Pu et al., 2022; Hua et al., 2023; Koptev and Cloetingh, 2024), facilitating magma resurgence zones and exerting a substantial influence on the evolution of rifting. Upon reaching the continental lithosphere, the melts could have additionally propagated over considerable distances through magma plumbing systems, thereby contributing to the structural net of dike swarms across extensive areas of West

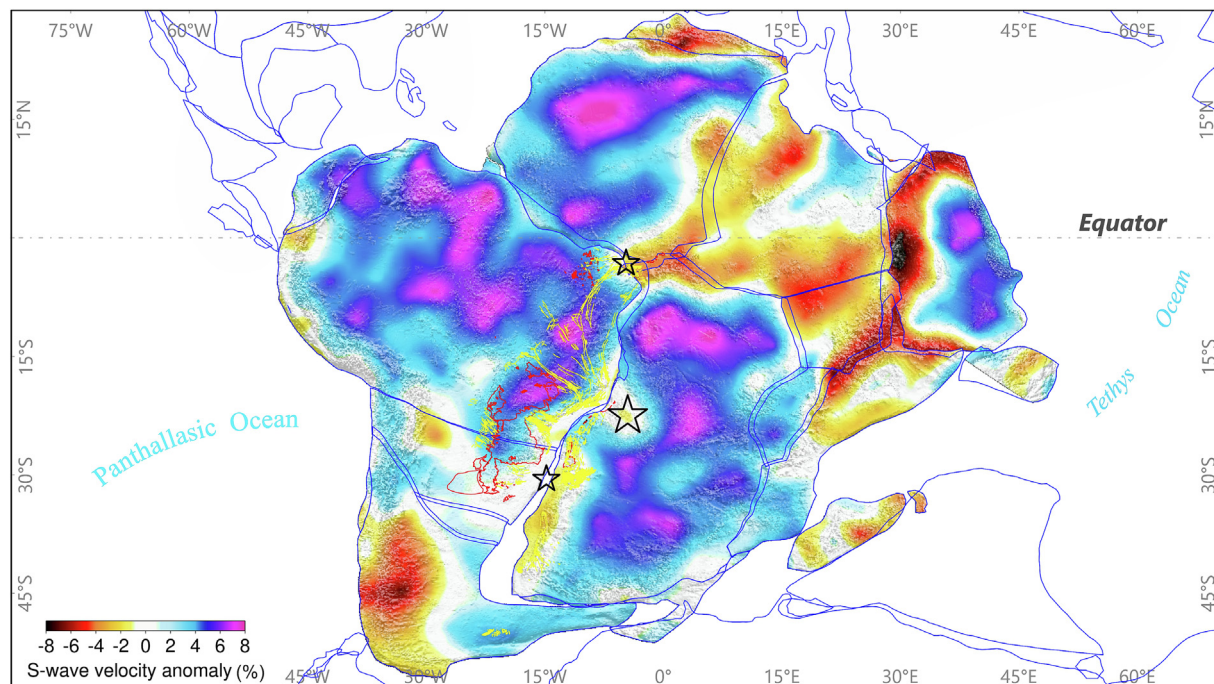


Fig. 13. Reconstruction of the West Gondwana at 134 Ma (Scotese, 2016). S-wave velocity variations at 110 km depth from the SA2019 tomography model (Celli et al., 2020). Dikes, basalt flows, and sills from the SAMP are superimposed (Macêdo Filho et al., 2023a). Note the possible magmatic centers coinciding with domal lithospheric structures in West Gondwana (black stars), from which large-scale mafic dike systems appear to radiate.

Gondwana – a pattern equally observed in other LIPs (e.g., Hastie et al., 2014; Ernst et al., 2019; Elliot and Fleming, 2021; Ben Marzoug et al., 2024; El Bilali and Ernst, 2024; Liang et al., 2024).

Recently, Sampaio et al. (2024) proposed that widespread EM signatures across the South Atlantic/Indian Oceans may reflect lithospheric recycling by mantle plumes. Within this framework, it is plausible that fragments of the ancient enriched continental lithosphere were earlier subducted and stored in deep mantle reservoirs and resurfaced via mantle(s) plume(s) in the Early Cretaceous. Upon impinging on the base of the SCLM, the plume may have induced additional melting of the overlying layer, generating hybrid liquids with trace element content broadly comparable to OIBs, yet still preserving subtle lithospheric geochemical features. In this context, the persistence of an EMI flavor may reflect the lack of isotopic gradients between the main contributing mantle reservoirs.

5.5. Boundaries of Atlantic LIPs in South America

In South America, the CAMP is prominently recognized across the north-central region of the continent, spanning the Guiana Shield, Amazonian Craton, and several sedimentary basins (Solimões, Amazonas, Acre, Alto Tapajós, Marajó, Parecis, Parnaíba, and Lavras da Mangabeira), along with the Precambrian Borborema and Tocantins provinces (Marzoli, 1999; Marzoli et al., 2018). More limited occurrences of CAMP are also reported in Bolivia. The magmatic records of the SAMP are concentrated in the Precambrian terrains of the Mantiqueira, Tocantins, and Borborema provinces, as well as within the intracratonic Paraná and Parnaíba basins (Macêdo Filho et al., 2023a,b,c). In the Parecis Basin (central-western Brazil), CAMP is associated with the Anari/Tapirapuã Formation along the southwestern margin of the basin (Waichel et al., 2024), whereas minor Early Cretaceous basalts occur on its eastern side (Rubert et al., 2019).

The lithospheric discontinuities of South America appear to have played a role in the emplacement of Mesozoic LIPs (Fig. 14).

The Az-125 (Alto Parnaíba) Lineament seems to have acted as an accommodation zone for the intrusion of high-Ti tholeiitic dikes of the Transminas swarm in the Early Cretaceous (Fig. 14). This lithospheric discontinuity also facilitated the ascent and emplacement of Late Cretaceous/Tertiary alkaline intrusions, carbonatites, and kimberlites. According to Stotz et al. (2025), dike swarms may reduce lithospheric mechanical strength, generating structural weaknesses that localize extensional deformation. Therefore, it is possible that the Az-125 swarm acted to concentrate younger intrusions along the lineament. Moreover, in parts of Central Brazil, Proterozoic mafic-ultramafic intrusions also appear to be linked to this structure (Moraes Rocha et al., 2014). If this older association is confirmed, the lineament may represent a long-lived, lithospheric-scale weakness zone that has channeled mantle-derived magmas multiple times throughout geological history. It is still noteworthy that the Alto Parnaíba Lineament effectively delineates the style of the SAMP magmatic remnants, with the southern block displaying both intrusive and extrusive magmatism, in contrast to the exclusively intrusive character of the northern block (Fig. 14).

The Proterozoic shear zone forming the Transbrasiliano Lineament appears to have acted as a significant tectonic barrier in West Gondwana, effectively separating the influence zones of CAMP and SAMP. One piece of evidence for this is the diffuse pattern acquired by the Az-125 dikes as the swarm approaches the megastructure (Fig. 14). In northeastern Brazil, the Parnaíba Basin covers the Transbrasiliano Lineament, thereby preserving both magmatic events: Triassic-Jurassic flood basalts are dominant on the western flank of the basin, while Early Cretaceous intrusions are more prominent on its eastern side. However, CAMP intrusions have also been reported along the eastern border of the basin near the Transbrasiliano Lineament (Macêdo Filho et al., 2023b). A comparable pattern is observed in Africa, where Chabou et al. (2010) identified the northern extension of the Transbrasiliano-Kandi Lineament (separating the West African Craton from the Pan-African Domain) as the eastern boundary of the CAMP in southwestern Algeria. In

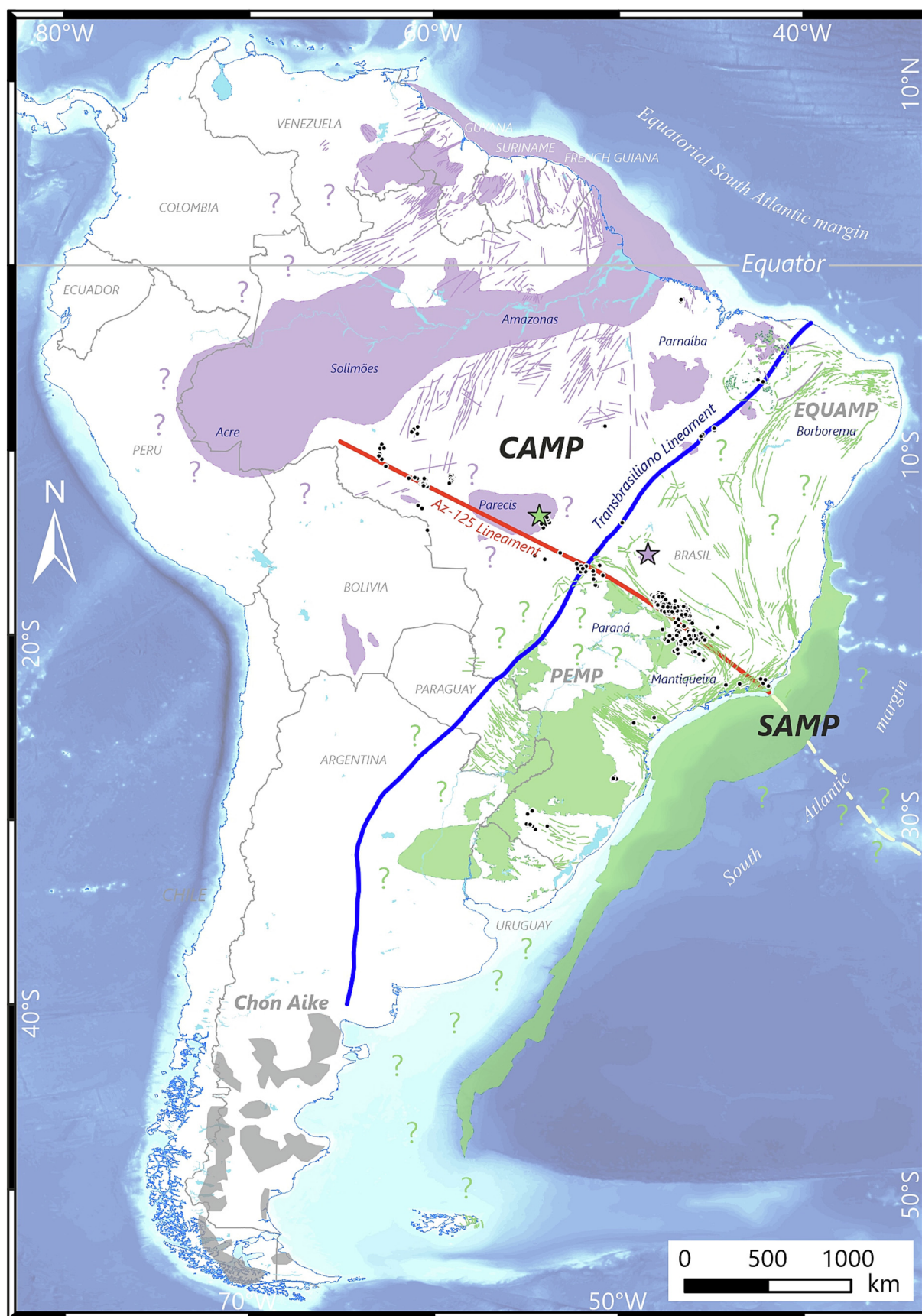


Fig. 14. Distribution of tholeiitic magmatic events associated with the fragmentation of the supercontinents Pangea (Triassic) and West Gondwana (Early Cretaceous) in South America. The Transbrasiliano Lineament serves as a natural geological boundary, separating the domains of magmatic events related to the Pangea breakup (CAMP) and the West Gondwana breakup (SAMP). The green star represents the Santiago do Norte basalts, occurring in the CAMP realm, while the purple star represents CAMP dikes in the Pirenópolis region, located in the SAMP realm. The black dots represent Mesozoic-Cenozoic alkaline, carbonatitic, and kimberlitic intrusions. Note the preferential occurrences along the Az-125 (Alto Paranaíba) Lineament. Abbreviations: CAMP, Central Atlantic Magmatic Province; SAMP, South Atlantic Magmatic Province. The question marks on the map indicate possible extensions of the LIP domain area.

the Gulf of Guinea region, CAMP remnants are found predominantly west of the Transbrasiliano Lineament, whereas only a limited number of dikes have been documented to the east (Tchouankoue et al., 2014).

In the southern Borborema Province, the Riacho do Cordeiro dike swarm and the Tucano-Reconcavo-Jatobá rift system seem to have been aborted due to the influence of the E-W-trending Proterozoic block delineated by the Pernambuco and Patos shear zones, which acted as a shield, preventing the further propagation of brittle deformation in the region (Macêdo Filho et al., 2023c; Dantas et al., 2024). In the northern Borborema Province, the Senador Pompeu dikes (CAMP; Oliveira et al., 2023) align with the Rio Ceará-Mirim dikes, both following the NE-SW structural trend of regional shear zones, suggesting some control of these lithospheric discontinuities in the emplacement/orientation of magma bodies (Macêdo Filho and Hollanda, 2022; Macêdo Filho et al., 2023c).

In central Brazil, CAMP and Neoproterozoic dikes align with the N-S ductile fabric of the Araguaia Belt (Cruz and Gorayeb, 2020), sharing a similar orientation with the N-S to NNW-SSE trending CAMP dikes further north in Amapá, Brazil (Sial et al., 1987). In the Carajás region (Amazonian Craton), examples of polycyclic intrusive events are documented, where Proterozoic, Cambrian, and CAMP dikes coexist within the same geological framework (Teixeira et al., 2019). These observations suggest that contrasting magmatic events, exploiting rheologically weakened domains, may have similarly influenced the Pirenópolis area, where CAMP and SAMP dikes were emplaced along NNW orientation. Collectively, the CAMP dikes from the Pirenópolis, Araguaia, Carajás, and Cassiporé regions form remnant pieces of a larger tectonic-magmatic puzzle, all converging toward the focal zone of CAMP magmatism during the early stages of Pangea breakup.

6. Conclusion

Our integrated dataset, including new whole-rock geochemical analyses, Sr-Nd-Pb isotopic compositions, mineral chemistry, $^{40}\text{Ar}/^{39}\text{Ar}$ and K-Ar geochronology, provided new insights into the petrogenesis and provinciality of Mesozoic dike swarms in central and southeastern Brazil.

The low-Ti dikes of the Pirenópolis swarm exhibit both compositional and temporal correlations with the Triassic magmatism (~210.7–205.5 Ma) of the Central Atlantic Magmatic Province. These newly identified CAMP dikes in central Brazil, together with recently reported occurrences from northeastern Brazil and Nigeria, substantially expand the known geographic extent of this LIP. Our revised estimate suggests a total area of $\sim 1.4 \times 10^7 \text{ km}^2$ across the Pangea landscapes, reaffirming CAMP as one of the largest and most significant LIPs in Earth's history.

The high-Ti dikes of the Transminas and Pirenópolis display geochemical affinity with Early Cretaceous magmatism (~133–131.5 Ma) of the Paraná-Etendeka and EQUAMP. The inter-LIP correlation is further supported by Ar-based geochronology and recent U–Pb (ID-TIMS) dates from various lithologies of the Paraná, Etendeka, and Equatorial Atlantic provinces (Rocha et al., 2020; Kingsbury et al., 2023; Sun et al., 2024, 2025; Oliveira et al., 2025). Taken together, these results support the interpretation that Early Cretaceous tholeiitic magmatism in West Gondwana represents a single, first-order LIP, here referred to as the South Atlantic Magmatic Province (SAMP).

The Azimuth 125° Lineament is interpreted as a lithospheric-scale structure that underwent at least two episodes of tectono-magmatic activity during the Phanerozoic, including the emplacement of the Transminas dikes during the Early Cretaceous and the intrusion of Late Cretaceous alkaline, carbonatitic, and kimberlitic bodies. We also emphasize the pivotal role of the Transbrasiliano

Lineament as a major lithospheric boundary that separated distinct magmatic and tectonic regimes associated with the fragmentation of Pangea (CAMP) and West Gondwana (SAMP).

Data availability

The data that support the findings of this study are openly available in the DIGIS Geochemical Data Repository hosted by GFZ Data Services (Macêdo Filho et al., 2026b) as well as in the [online supplementary material](#).

CRediT authorship contribution statement

Antomat A. Macêdo Filho: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Valdecir A. Janasi:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis. **Alisson L. Oliveira:** Writing – review & editing, Software, Methodology, Formal analysis, Data curation. **Maria Helena Hollanda:** Writing – review & editing, Resources. **Dawid Szymanowski:** Writing – review & editing, Visualization, Validation, Formal analysis. **Elton L. Dantas:** Writing – review & editing, Resources, Data curation. **Richard E. Ernst:** Writing – review & editing. **Alexandre O. Chaves:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at <https://doi.org/10.1016/j.gsf.2026.102306>.

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